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资本市场自由化是否推动数字化转型？来自中国 A 股纳入 MSCI 新兴市场指数的证据

Does Capital Market Liberalization Drive Digital Transformation? Evidence from Chinese A-Shares' Inclusion in the MSCI Emerging Markets Index

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摘要

本文聚焦新兴市场情境下资本市场自由化能否因果性推动企业数字化转型。本文以中国 A 股分阶段纳入 MSCI 新兴市场指数为制度性冲击，该冲击提升了样本企业面向全球投资者的可得性与关注度，构成具有政策含义的准自然实验。基于 2010–2022 年 2,712 家上市公司、35,264 个公司-年份观测值，本文采用双向固定效应双重差分（DID）识别策略，并结合事件研究、倾向得分匹配双重差分（PSM-DID）、安慰剂检验与分期处理诊断开展稳健性验证。

本文的核心贡献之一是构建了基于中文年报文本的企业数字化转型测度指标。该指标采用自然语言处理与领域词典方法，覆盖人工智能、大数据、云计算、物联网、区块链/分布式账本与企业数字化推进六类技术集群。为缓解关键词膨胀与战略性披露偏差，本文将文本指标与外部技术相关结果进行效度检验，并结合对数变换、缩尾处理、披露长度标准化指标及替代指标设定进行敏感性分析。

实证结果显示，在优选基准模型下，MSCI 纳入使企业数字化转型强度提高 0.39 个对数点，约相当于处理前水平提升 48%。动态事件研究结果表明，处理前不存在系统性趋势差异，处理后效应持续且具有累积特征。进一步地，安慰剂处理、分期处理诊断、替代聚类标准误、受限样本检验与多口径被解释变量结果均支持核心结论。异质性分析发现，该效应在高技术行业和经济较发达地区更为显著，表明吸收能力与制度互补性会影响自由化收益。机制检验对两条可直接识别的渠道提供一致证据：融资约束缓解，以及外部监督与信息披露纪律强化带来的治理改善。总体而言，本文表明国际金融一体化可作为新兴市场企业数字化升级的重要制度催化机制。

关键词：资本市场自由化；数字化转型；MSCI 纳入；中国 A 股；新兴市场；计算文本分析；双重差分

Abstract

This thesis investigates whether capital market liberalization can causally promote digital transformation at the firm level within an emerging-market context. The empirical setting involves the phased inclusion of Chinese A-share firms into the MSCI Emerging Markets Index, serving as a policy-relevant institutional shock that increased the exposure of selected firms to global investors. Building upon this quasi-natural experiment, the study integrates an identification-oriented econometric methodology with comprehensive computational measurement to generate evidence that is both causally interpretable and substantively significant. Specifically, the analysis employs a panel comprising 35,264 firm-year observations from 2,712 listed firms spanning the years 2010 to 2022 and estimates treatment effects utilizing a two-way fixed-effects difference-in-differences framework, complemented by event-study dynamics, matching-based diagnostics, and placebo-based falsification.

A central contribution of this thesis is the development of a comprehensive digital transformation outcome metric derived from Chinese annual report text. The metric is constructed employing natural language processing techniques and domain-specific dictionaries encompassing six technology clusters: artificial intelligence, big data analytics, cloud computing, Internet of Things, blockchain/distributed ledger technologies, and enterprise digitization initiatives. To mitigate concerns regarding keyword inflation and disclosure bias, the study validates the text-based indicators against external technology-related outcomes, applies logarithmic transformations and winsorization procedures, and conducts robustness tests using length-normalized outcomes and alternative index formulations.

The baseline results demonstrate that MSCI inclusion enhances digital transformation intensity by 0.39 log points within the preferred specification, which is approximately a 48% increase relative to pre-treatment levels. Dynamic event-study estimates show no systematic divergence prior to treatment and indicate enduring post-treatment effects, thereby supporting the parallel trends assumption and suggesting a cumulative adjustment process rather than a one-period reporting response. An extensive array of robustness checks, including placebo treatment assignments, staggered adoption di-

agnostics, propensity-score-matched DID estimation, alternative clustering strategies, restricted-sample tests, and multiple dependent-variable definitions, substantiate this conclusion.

The thesis further explores heterogeneity and mechanistic pathways to transcend average effects. Heterogeneity analyses indicate more pronounced effects in high-technology industries and economically advanced regions, suggesting that absorptive capacity and institutional complementarity influence the benefits of capital market liberalization. Evidence of mechanisms consistently supports two directly testable channels: alleviation of financing constraints and enhancement of governance quality through intensified external oversight and disclosure discipline. Overall, the study demonstrates that international financial integration can serve as an institutional catalyst for corporate digital upgrading in emerging markets.

Keywords: Capital market liberalization; Digital transformation; MSCI inclusion; Chinese A-shares; Emerging markets; Computational text analysis; Difference-in-differences

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List of Abbreviations

Abbreviation	Full Term
AI	Artificial Intelligence
ATT	Average Treatment Effect on the Treated
C1	Basic Manufacturing (CSRC Subcategory Code)
C2	Equipment Manufacturing (CSRC Subcategory Code)
C3	High-End Manufacturing (CSRC Subcategory Code)
C27	Transportation Equipment Industry (CSRC Industry Code)
C38	Electrical Equipment Industry (CSRC Industry Code)
C39	Instrumentation Industry (CSRC Industry Code)
CF	Cash Flow
CNINFO	China Securities Information Disclosure Platform
CNRDS	China Research Data Service
CNY	Chinese Yuan
CSMAR	China Stock Market and Accounting Research Database
CSRC	China Securities Regulatory Commission
DID	Difference-in-Differences
DOI	Digital Object Identifier
DT	Digital Transformation
DT_Comp	Composite Digital Transformation Index
EM	Emerging Markets
ERP	Enterprise Resource Planning
FE	Fixed Effects
GTA	GTA Information Technology Co.
H1	Hypothesis 1
H2	Hypothesis 2
H3	Hypothesis 3
HHI	Herfindahl-Hirschman Index
HMM	Hidden Markov Model
IaaS	Infrastructure as a Service
ICT	Information and Communication Technology
IMF	International Monetary Fund

Abbreviation	Full Term
IoT	Internet of Things
IP	Intellectual Property
IT	Information Technology
KZ Index	Kaplan-Zingales Financing Constraint Index
MSCI	Morgan Stanley Capital International
NLP	Natural Language Processing
NPV	Net Present Value
PaaS	Platform as a Service
PSM	Propensity Score Matching
PSM-DID	Propensity Score Matching Difference-in-Differences
QFII	Qualified Foreign Institutional Investor
R2	Coefficient of Determination
RMB	Renminbi
ROA	Return on Assets
ROE	Return on Equity
RQFII	RMB Qualified Foreign Institutional Investor
SaaS	Software as a Service
SD	Standard Deviation
SE	Standard Error
SIC	Standard Industrial Classification
SME	Small and Medium-Sized Enterprise
STAR	Science and Technology Innovation Board
TWFE	Two-Way Fixed Effects

Chapter 1 Introduction

This chapter introduces the core research problem, motivates its academic and policy relevance, and defines the thesis objectives and scope. It positions the study at the intersection of capital market liberalization and firm-level digital transformation, clarifies why the MSCI A-share inclusion episode provides a suitable quasi-experimental setting, and outlines the empirical and methodological contributions developed in subsequent chapters.

Beyond orientation, the chapter specifies the analytical stance adopted throughout the thesis: digital transformation is treated as an investment process shaped not only by internal managerial capability but also by external capital-market institutions. This framing shifts the research question from *whether* firms transform to *how* financial opening changes the conditions under which transformation becomes feasible, credible, and sustained.

1.1 Research Background

The relationship between capital market liberalization and corporate digital transformation is currently a hot topic, especially in development economics, corporate finance, and technology management. Over the last two decades, governments in emerging nations have gradually opened their domestic financial institutions to international investors. At the same time, firms have been under increasing pressure to digitize production, operations, governance, and customer interactions. Despite the importance of these themes, prior research has often explored them separately rather than together. Financial liberalization research typically focuses on valuation implications, liquidity, and investment efficiency, whereas digital transformation research focuses on internal skills, leadership, and technology readiness[1]. The critical question at the intersection of these literatures remains unanswered: can international capital market integration promote firm-level digital transformation, and if so, through which mechanisms?

This thesis is driven by the fundamental question concerning boundaries. It states that digital transformation involves both a financing and governance process, shaped by

capital market institutions, as well as a management or technological process. Characteristics of digital transformation programs in many emerging markets include high fixed costs, uncertain returns, prolonged gestation periods, and strong complementarities with management learning[2]. Even when expected social benefits are substantial, these features may introduce funding frictions that potentially hinder or delay adoption[3]. Capital markets liberalization might alter the investment considerations of firms contemplating digital upgrades by expanding the investor base, decreasing financing costs, and improving governance[4]. Gaining a comprehensive understanding of this connection is essential for both scholarly inquiry and policy making.

The Chinese background is especially important for this research. China features scale, diversity, and institutional growth in ways that make it ideal for empirical study[5]. It is one of the world's largest stock markets and a leading hub for industrial digital transformation, yet capital account liberalization has mostly been gradual and driven by policy[6]. This slow process results in notable policy events with predictable timing. Among these, the inclusion of Chinese A-share firms in the MSCI Emerging Markets Index stands out as a key externally imposed reform that increased firms' exposure to international investors.

The practical reason behind this study is equally strong. Policymakers increasingly view digital transformation as a way to boost productivity and national competitiveness. However, firms vary greatly in their capacity to carry out transformation initiatives[7]. Market-opening policies may produce wider technical spillovers than are typically acknowledged if access to global capital and investor networks significantly influences digital adoption[8]. On the other hand, liberalization may increase regional and firm-level disparities in digital capabilities if gains are concentrated among firms that already have advantages. Therefore, a thorough empirical evaluation is required.

Another important motivator is the measurement technique. A large portion of the research on digital transformation is based on managerial evaluations, survey data, or limited proxies like IT spending and patent counts. These metrics frequently have poor comparability over time, self-reporting bias, and restricted coverage[9]. In order to create a firm-year digital transformation index from annual report disclosures, this thesis uses a

computational text-analysis approach. The project aims to generate evidence that is both strong in identification and wide in coverage by combining this large-scale measuring technique with a quasi-experimental methodology.

Chen et al.[10] show that increased analyst coverage and improved information accuracy are two mechanisms through which digital transformation strengthens firms' information environments. Cheng and Masron[11] further argue that regulatory uncertainty can intensify firms' digital responses, highlighting the central role of external institutional conditions in shaping transformation behavior. Du and Wang[12] find that the innovation benefits of digital transformation are significantly amplified by financial support, particularly through capital markets, with effects significant at the 1% level. Zhang and Wang[13] report that digitally transformed firms face lower equity financing costs, whereas Xue and Zhang[14] document improvements in debt-financing outcomes and overall firm performance. Taken together, this evidence suggests that the financial consequences of digital transformation are conditional on financing environment, disclosure quality, and governance structure.

While moving beyond correlation-based inference, this study builds on previous research. Most earlier studies focus on a single channel or outcome variable, often without addressing treatment assignment endogeneity. In contrast, this thesis combines mechanism-oriented research, a long panel (2010–2022), and a quasi-experimental policy shock to evaluate whether liberalization can be considered a causal factor in the level of digital transformation. The theory conceptualizes digital transformation as a process of cumulative strategic investment. Over time, firms adjust governance procedures, assess financing constraints, and update expectations. At least three mechanisms are expected for liberalization to influence this process: removing financial restrictions, enhancing governance and transparency standards through international oversight, and expanding access to cross-border knowledge networks. These mechanisms form the foundation of the thesis's empirical framework.

As a result, there are three main motivations. First, it tackles a significant open question in the literature about the relationship between firm-level technological innovation and global finance. Secondly, it offers evidence from a top and emerging economy that

is methodologically robust. Third, it provides firms and regulators looking to integrate capital market reform with digital development goals with policy-relevant insight.

1.2 Research Questions and Objectives

In light of the aforementioned motivation, this thesis addresses three relevant research problems.

- First, does inclusion in the MSCI Emerging Markets Index, a proxy for capital market liberalization, significantly increase firm-level digital transformation intensity? This is the thesis's primary causal question. Specifically, the study evaluates whether treated firms exhibit a sustained post-inclusion increase in digital transformation relative to otherwise comparable untreated firms.
- Second, do treatment effects vary by technology domain, industry, area, and firm characteristics? Digital transformation is not a single technology, but a collection of investments, each with its own capital intensity and capabilities requirements. The thesis investigates if effects are stronger in domains such as AI, big data, and cloud computing; whether reactions differ across technology-intensive and traditional industries; and whether regional institutional quality influences outcomes.
- Third, how does liberalization affect digital transformation? The thesis investigates three channels using theory and prior evidence: funding restrictions, governance and monitoring, and knowledge spillovers. In empirical implementation, this thesis directly tests the first two channels and treats the knowledge channel as a deferred empirical extension requiring direct mediator variables.

The methodological objective is to combine proven NLP-based measurement with quasi-experimental econometrics in a transparent, reproducible, and appropriate manner for large-scale panel data. The key objective is to determine whether international financial integration may be used as a stimulus for technological advancement within firms. The policy objective is to offer information to capital market regulators, institutional investors, and corporate executives about the larger technology implications of market-opening changes.

The thesis follows five operational aims to achieve these objectives. It generates high-coverage firm-year digital transformation metrics from annual reports. It employs baseline and dynamic DID techniques to estimate treatment effects. It evaluates inferential stability through robustness and sensitivity tests. It uses heterogeneous effects to identify boundary conditions. It conducts mechanism evaluations to assess financing, governance, and knowledge channels.

1.3 The Empirical Setting: China's A-Share Market and MSCI Index Inclusion

The empirical background for this argument is the gradual inclusion of Chinese A-shares in the MSCI Emerging Markets Index, a watershed moment in China's financial opening process[15]. Prior to 2018, global investors had limited access to China's domestic equity market due to ownership restrictions, quota systems, and market-access frictions. Although conduits such as QFII, RQFII, and the Stock Connect programs existed, broad passive global allocation to mainland A-shares remained limited[16].

MSCI inclusion changed this environment by connecting eligible A-share companies to benchmark-driven international capital flows[17]. MSCI indexes are often used by global institutional investors and index-tracking funds. Being included in the MSCI Emerging Markets Index affects active portfolio building, analyst focus, and investor relations practices, in addition to passive investment flows[6]. The reform therefore modified firms' exposure to external financing and oversight conditions in a way that is economically significant.

The phased inclusion process is particularly useful for identification. Phase I was implemented in June 2018 with 234 large-cap A-shares at a 5% inclusion factor following the June 2017 announcement. By November 2019, Phase II expanded coverage to 472 securities with a higher inclusion factor. This phased implementation provides clear treatment timing and observable pre- and post-policy windows. A key advantage of this setting is that inclusion criteria were largely rule-based and externally determined. Eligibility was based on market capitalization, liquidity, and Stock Connect accessibility rather than on discretionary judgments about firms' digital strategies[18]. Although no

real-world setting is fully random, this institutional design strengthens the plausibility that inclusion captures exogenous variation in international capital access relative to firms' underlying digital transformation trajectories.

China is also an ideal context because firms operate under substantial heterogeneity in regional development, industry composition, ownership structures, and governance quality. This heterogeneity allows the thesis to test whether liberalization effects are uniform or conditional on absorptive capacity[19]. Zhang et al.[20] show that treatment effects vary systematically with regional market development, ownership structure, and firm size, with stronger responses in certain non-Eastern and non-state-firm contexts. Embedding these moderation patterns within the MSCI setting helps move beyond average treatment effects toward a richer understanding of distributional outcomes.

1.4 Overview of Research Design and Data

The empirical design combines causal identification with large-scale computational measurement. The main approach is a two-way fixed effects difference-in-differences model comparing firms that are included in MSCI (treated) and those that are not (untreated) before and after inclusion. This baseline is supplemented by event-study estimation to evaluate dynamic responses and pre-trend validity, propensity score matching DID to enhance comparability on observables, and placebo-based inference checks.

The data cover 35,264 firm-year observations from 2010 to 2022, representing 2,712 Chinese listed firms after sample cleaning. Core financial and governance variables are drawn from CSMAR, while treatment status is derived from MSCI constituent lists. Digital transformation outcomes are constructed from annual-report text using NLP-based keyword extraction across six technology domains: AI/machine learning, big data, cloud computing, IoT, blockchain/distributed ledger technologies, and enterprise digitization programs.

The baseline outcome is a composite index using log transformation of aggregated keyword frequencies. This approach addresses right skewness and enhances interpretability when disclosure dispersion is high. For robustness, alternative outcomes are also used, including domain-specific indices, normalized measures adjusted for report

length, and additional transformations to assess sensitivity to scaling choices.

Mechanism analysis is embedded in the design rather than added as a descriptive task. Proxies related to financing constraints and governance quality are estimated through structured regressions to check whether channel patterns match the conceptual framework. The knowledge-spillover channel is retained at the theoretical level but not directly estimated in the baseline because direct mediator variables are not yet available in the merged panel.

Overall, the research design is deliberately triangulated: one treatment setting, multiple estimators, multiple outcome measures, and various mechanism tests. This triangulation enhances confidence that the results reflect genuine economic effects rather than model-specific artifacts.

1.5 Significance of Research

This study is significant for three strands of literature: capital market liberalization, digital transformation economics, and computational empirical methods.

- First, it expands the liberalization literature beyond traditional market results. Previous research often focuses on liquidity, valuation, return co-movement, or financing costs. This study demonstrates that liberalization can affect firms' technological paths, thus connecting financial integration to long-term productivity-related behavior.
- Second, it advances the digital transformation literature by highlighting external financial environments as a key factor in shaping corporate digital strategy. While much existing research focuses on internal capabilities, managerial cognition, and organizational change, this thesis demonstrates that access to international capital and investor oversight can significantly influence the speed and direction of transformation.
- Third, it offers mechanism-level evidence that directly tests financing and governance channels under a common treatment design, while preserving knowledge spillovers as a theoretically grounded channel for future direct testing.
- Fourth, the thesis enhances methodology by combining quasi-experimental causal

inference with validated Chinese-language text analytics at scale.

- Fifth, it contributes to the emerging-market comparative literature by offering high-resolution evidence from China.
- Sixth, it provides policy implications based on micro-level evidence. Regulators interested in the broader development impact of capital market reform can use these findings to assess whether market-opening policies promote strategic upgrading.

1.6 Thesis Structure

The thesis is organized in seven chapters, with each chapter designed to build logically on the previous one.

- **Chapter 1** introduces the research problem, motivation, questions, objectives, empirical setting, and expected contributions.
- **Chapter 2** develops the literature review. It synthesizes foundational and recent studies on liberalization, digital transformation, mechanism channels, and text-based measurement.
- **Chapter 3** presents institutional background and formal hypothesis development. It details China's pre-liberalization capital market structure, the MSCI inclusion process, and the quasi-experimental properties of the reform.
- **Chapter 4** introduces Research Design I focused on causal identification. It describes sample construction, variable definitions, treatment assignment, baseline and dynamic model specifications, and descriptive statistics.
- **Chapter 5** introduces Research Design II focused on computational measurement. It documents corpus construction, Chinese text segmentation, dictionary design, index construction, and validity diagnostics.
- **Chapter 6** reports empirical results and discussion. It presents baseline effects, event-study dynamics, robustness checks, heterogeneity analyses, and mechanism evidence.
- **Chapter 7** concludes with a synthesis of findings, original contributions, limitations, and future research directions.

Chapter 2 Literature Review

This chapter builds the theoretical and empirical foundation for the study by synthesizing research on capital market liberalization, digital transformation, and computational measurement. It examines the main mechanisms through which access to international capital may impact firm behavior, assesses existing evidence on contextual differences, and highlights unresolved gaps in identification and mechanism integration.

2.1 Capital Market Liberalization: Theory and Evidence

2.1.1 Theoretical Foundations of Financial Liberalization

Capital market liberalization generally refers to removing barriers that separate domestic financial systems from international portfolio flows[21]. In essence, liberalizing equity markets can improve allocative efficiency by enabling capital transfer to higher-productivity companies, lowering financing costs through more investor participation, and increasing risk-sharing via diversified ownership structures[22]. The traditional welfare argument suggests that firms facing financial constraints should benefit more when foreign capital is available[23]. In segmented markets, limitations of domestic savings, home bias, and informational frictions can sustain high costs linked to external finance. Liberalization efforts can reduce these frictions by boosting competition among investors and increasing market liquidity[24].

Nevertheless, theoretical frameworks also highlight non-linearities. The benefits linked to liberalization depend on the quality of institutions, the effectiveness of legal enforcement, disclosure standards, and the strength of market infrastructure[25]. In environments with weak institutional frameworks, liberalization may increase market volatility or shift resources toward short-term opportunities instead of encouraging productive long-term investments[26]. From a corporate perspective, liberalization can be seen as a change in the optimization problems facing managers and controlling shareholders[11]. It affects the relative cost of external financing, the expected investment horizon of investors, and the level of monitoring for capital provision[27].

The MSCI inclusion mechanism is particularly informative because it combines

signaling and flow effects. Inclusion can automatically trigger passive fund allocations while simultaneously signaling market quality and investability to active global investors.

2.1.2 Effects on Corporate Investment, Innovation, and Governance

Empirical studies generally find that market opening influences corporate behavior through investment and governance channels[28]. On the investment side, reduced financing frictions can broaden firms' feasible project options[29]. Firms may increase R&D, capital expenditure, and strategic intangible investments when expected financing constraints lessen[23]. On the governance side, foreign institutional investors can strengthen monitoring and raise transparency expectations.

Evidence from prior work supports this logic. Chen et al.[10] show that digital transformation is associated with higher analyst coverage and improved information accuracy. Du and Wang[12] provide direct evidence that financial support strengthens the innovation effects of digital transformation, particularly in environments with tighter conventional financing constraints. Chen and Zhang[3] and Feng et al.[30] extend this argument to firms' financing structures.

2.1.3 Evidence from Emerging Market Contexts

Emerging markets provide valuable evidence because they combine high growth potential with institutional frictions that highlight financing channels. Studies in these markets often detect stronger real effects of financial reforms compared to more developed markets where constraints are already less severe[31]. The Chinese evidence is especially useful due to detailed data and specific policy information[32]. However, a direct causal link from liberalization shocks to firm-level digital transformation outcomes is less developed.

Zhang et al.[20] show that contextual heterogeneity is central: estimated effects vary with regional market development, ownership structure, and firm size. This pattern implies that pooled average treatment estimates may mask economically meaningful distributional differences. Stiglitz[33] similarly emphasizes that policy shocks interact with organizational absorptive capacity.

2.2 Digital Transformation: Conceptual Framework and Determinants

2.2.1 Defining Digital Transformation: Technological and Organizational Dimensions

Digital transformation extends beyond just acquiring technology. It involves continuous, firm-wide reconfiguration of value creation, operational processes, and organizational coordination through digital technologies[34]. A narrow view that only considers software or hardware spending misses the strategic and organizational aspects that influence whether digital investments lead to productivity improvements. Conceptually, digital transformation includes at least three layers[14,35]: infrastructure adoption, process transformation, and organizational transformation.

These layers depend on each other. Infrastructure without process integration can yield limited results[36]. Process changes without organizational incentives can prevent progress. Organizational commitment without funding can remain superficial[14]. Therefore, digital transformation is best viewed as a coordinated investment journey rather than a single technological event[2].

2.2.2 Core Technology Domains: AI, Big Data, Cloud Computing, IoT, and Blockchain

The technology areas highlighted in this thesis align with key trends in modern corporate digital transformation. Artificial intelligence and machine learning support prediction, optimization, and automation across both front-end and back-end operations[37]. Big data systems enable companies to turn large, fast-moving information into operational insights. Cloud computing offers scalable infrastructure that reduces fixed ICT costs[38]. Internet of Things technologies link physical assets to digital monitoring and control systems[39]. Blockchain and distributed ledger technology provide traceability, tamper resistance, and coordination across organizational boundaries[40]. Enterprise digitization initiatives combine these technologies with ERP upgrades, process engineering, and platform strategies.

These domains vary in capital intensity, knowledge requirements, and implementation risks[41]. Existing research supports this multidimensional view[42].

2.2.3 Drivers, Barriers, and Firm-Level Heterogeneity

Drivers of digital transformation include financial slack, managerial capability, data infrastructure, and complementary human capital. Barriers are also significant. High uncertainty about benefits can discourage investment, particularly in firms under pressure for short-term performance. Talent shortages in data engineering, AI deployment, and cybersecurity limit progress[43]. Legacy systems can increase integration costs.

These barriers imply heterogeneous response capacities. Larger firms may benefit from economies of scale in technology investment[44]. Companies in developed regions might access more extensive supplier networks and skilled labor. Non-state firms often face greater market competition, creating stronger incentives to adopt productivity-enhancing digital strategies[45].

2.2.4 Digital Transformation in Emerging Markets: Opportunities and Constraints

Emerging markets offer a unique digital transformation landscape with both growth opportunities and structural challenges[46]. In this setting, access to external capital markets can serve a catalytic function. International investors may not only supply funding but also bring governance oversight, benchmarking standards, and a focus on long-term capability building[47]. In China, the coexistence of advanced digital ecosystems and uneven regional development creates an especially revealing context for identifying heterogeneous treatment effects[48].

2.3 Mechanisms Linking Capital Market Liberalization to Digital Transformation

2.3.1 The Financing Constraint Channel

The financing-constraint mechanism is the most direct link between liberalization and digital transformation. Digital projects often require large upfront expenses in software systems, data architecture, cloud migration, machine learning platforms, and organizational training[49]. By broadening access to equity investors and lowering required returns, liberalization can reduce the user cost of capital for these projects[50].

Existing studies provide direct support for this channel. Hordofa[51] shows that capital-market financial support significantly strengthens the innovation effects of digital transformation, while Liu[15] documents lower equity financing costs among firms undergoing digital upgrading. Together, these findings suggest that financing and digital transformation are mutually reinforcing. The channel is unlikely to be homogeneous across firms: firms with initially binding financing constraints should exhibit larger responses[52].

2.3.2 Corporate Governance and International Investor Monitoring

A second mechanism is governance upgrading through international investor monitoring. Foreign institutional investors often demand higher disclosure quality, stronger board accountability, and clearer strategic communication[27]. Under these conditions, managers face stronger incentives to prioritize long-term value-enhancing projects and reduce agency-driven underinvestment.

Ma et al.[53] provide supporting evidence that digital transformation is associated with improved information environments through increased analyst coverage. Li[54] further shows that governance effects can interact with ownership structure.

2.3.3 Knowledge Spillovers and Global Best Practice Diffusion

A third mechanism operates through information and knowledge spillovers. Capital market liberalization boosts interactions between domestic firms and global investors, analysts, and intermediaries[8,22]. Through investor meetings, analyst reports, bench-

marking exercises, and governance engagement, firms can access implementation templates and performance standards[55]. This mechanism aligns with broader themes in the literature[24,56-57]. Evidence from competitive manufacturing environments further suggests that digital-technology spillovers can be strategically state-dependent across rival firms rather than mechanically uniform[58].

2.3.4 Market Signaling, Certification, and Reputational Effects

Liberalization can affect digital transformation through signaling and certification processes. Inclusion in a major global benchmark is often interpreted as recognition of minimum standards in accessibility, liquidity, and market relevance[54]. This signal can lower external uncertainty about firm quality. Dong[19] further distinguishes substantive transformation from symbolic signaling, an important distinction when interpreting post-inclusion disclosure behavior.

2.4 Text-Based Measurement of Firm Characteristics

2.4.1 Computational Approaches in Corporate Finance and Economics

The rise of text-as-data methods has revolutionized empirical corporate research by enabling high-frequency, large-sample analysis[59]. In the era of digital transformation, computational text methods are especially valuable because direct accounting measures are often incomplete[60]. Methodologically, text-based approaches enhance scalability and consistency compared to purely manual coding[61]. However, computational methods need careful validation[62].

2.4.2 NLP-Based Measurement of Innovation and Digital Activity

Afzal[63] shows that Chinese-language annual reports can provide credible NLP-based indicators of digital activity. Kumar[64] further argues for domain-cluster keyword extraction rather than generic buzzword counts. Li et al.[65] position NLP-based measurement as a complement to, rather than a substitute for, financial indicators.

2.4.3 Validation Strategies for Text-Derived Indices

Validation is central to any text-based empirical strategy. This thesis applies three validation layers: content validity, construct validity, and robustness validity. Content validity is established through careful dictionary design and expert review. Construct validity is evaluated by comparing text-derived indices with related observable variables. Robustness validity is assessed through alternative outcome definitions and algorithmic specifications.

2.4.4 Challenges in Chinese-Language Disclosure Environments

Chinese-language textual analysis presents technical and interpretive challenges[9,65]. Chinese text segmentation is complex because characters are written without spaces. Another challenge is variation in strategic disclosure[59]. A third challenge involves distinguishing between intention and implementation.

Despite these challenges, Chinese annual reports remain some of the most comprehensive and standardized data sources for long-term firm-level analysis. With careful design and validation, they offer a strong empirical foundation for studying digital transformation at scale.

2.5 Research Gaps and Positioning of This Study

The reviewed literature reveals four core gaps that motivate this thesis.

- **Linkage gap**: Research on capital market liberalization and research on digital transformation have mostly developed independently.
- **Identification gap**: Many studies rely on cross-sectional correlations or panel models that may face endogeneity issues.
- **Mechanism integration gap**: Existing evidence suggests financing, governance, and knowledge channels are all plausible but often examined in isolation.
- **Measurement scalability gap**: Direct digital investment metrics are often incomplete in emerging-market accounting systems.

This thesis addresses these gaps by examining the MSCI A-share inclusion event as a near-natural experiment to determine the causal effect of liberalization on digi-

tal transformation. The literature synthesis provides direct support for this positioning. What is missing is a design that integrates these insights into a single coherent causal framework. This thesis is constructed to provide that framework.

Chapter 3 Institutional Background and Hypothesis Development

This chapter establishes the institutional context and formal hypothesis structure underpinning the empirical analysis. It first explains the pre-liberalization structure of China's A-share market and the role of Stock Connect as implementation infrastructure for foreign participation. It then details the MSCI inclusion timeline and rule-based eligibility criteria, highlighting why the reform provides plausibly exogenous variation in international capital access. Finally, the chapter translates the conceptual framework from Chapter 2 into explicit testable hypotheses.

3.1 China's Capital Market Structure Prior to Liberalization

3.1.1 The A-Share Market and Domestic Investor Segmentation

China's A-share market—the renminbi-denominated equity market listed on the Shanghai and Shenzhen Stock Exchanges—was historically subject to strict foreign ownership restrictions and capital controls, which created near-complete segmentation from international capital flows[66]. Before liberalization reforms, foreign investors could access mainland Chinese equities only through the limited QFII (Qualified Foreign Institutional Investor) and RQFII programs, which imposed strict quotas and administrative barriers. This segmentation meant that Chinese listed firms relied almost entirely on domestic capital markets, characterized by shorter investment horizons, weaker corporate governance standards, and limited exposure to global best practices.

Figure 3-1 provides a data-based characterization of this segmented pre-liberalization structure using only pre-treatment observations (2010–2017). The chart reports standardized mean gaps between firms that are eventually included in MSCI (2018–2022) and firms never included. The gaps are economically large and statistically precise for firm size (1.20 standard deviations, $p < 0.01$), total market value (0.80 standard deviations, $p < 0.01$), and independent-director representation (0.47 standard deviations, $p < 0.01$), showing that the subset of firms later connected to global capital was already structurally distinct within the domestic A-share universe.

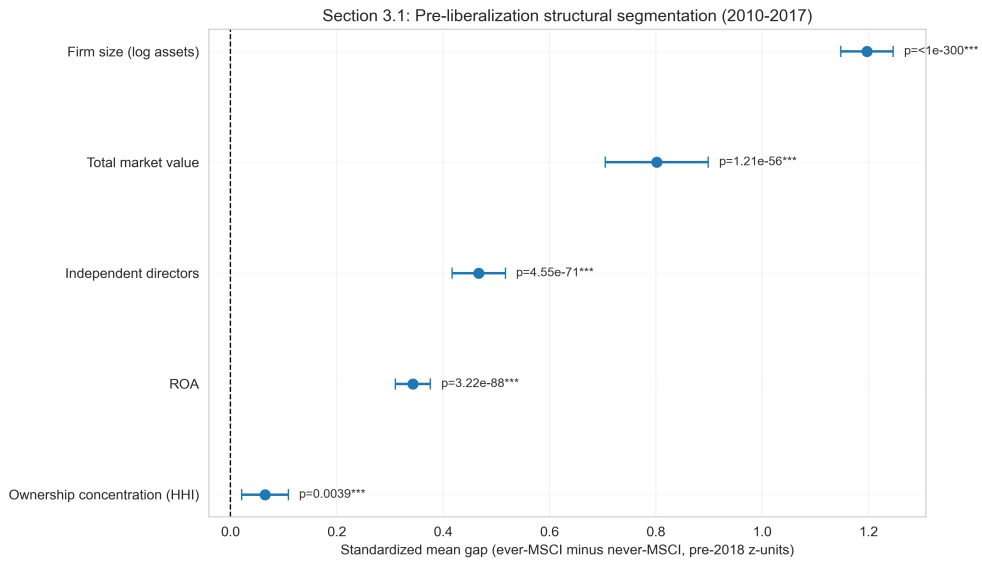

Figure 3-1 Section 3.1 Data Profile: Pre-Liberalization Structural Segmentation (2010–2017)

3.1.2 Stock Connect Programs as the Technical Infrastructure for Liberalization

The technical infrastructure for capital market liberalization was established through the Stock Connect programs: Shanghai–Hong Kong Stock Connect (launched November 2014) and Shenzhen–Hong Kong Stock Connect (launched December 2016)[67]. These programs developed daily and aggregate quotas, currency hedging methods, and settlement protocols that approved foreign investors to buy A-shares while adhering to capital control regimes. By 2018, these systems were sufficiently developed to handle the substantial capital movements associated with MSCI inclusion[23].

3.2 The MSCI Emerging Markets Index and the A-Share Inclusion Process

3.2.1 MSCI as a Global Benchmark: Scale, Influence, and Index Composition

The MSCI Emerging Markets Index is the main benchmark for global institutional investors investing in developing market equities, with about $1.9 trillion in assets under management tracking or comparing against MSCI emerging market indices in 2018. The index's composition determines mandatory portfolio allocations for passive funds

and acts as a reference for active managers, influencing capital flows to the included companies[17].

3.2.2 Inclusion Criteria: Market Capitalization, Liquidity, and Stock Connect Accessibility

MSCI's inclusion criteria were mechanical rather than discretionary: (i) a minimum free-float adjusted market capitalization threshold; (ii) minimum liquidity requirements based on annualized trading volume and turnover ratios; and (iii) accessibility through Stock Connect programs. Importantly, these criteria are independent of firms' underlying digital transformation paths or performance, offering a plausible source of exogenous variation in international capital access.

To make this point data-based, Figure 3-2 estimates a 2017 cross-sectional logit model where the dependent variable is eventual MSCI inclusion. The odds-ratio estimates show strong and statistically significant associations for each rule-based eligibility proxy: log total market value (OR = 17.04, $p < 0.01$), log free-float market value (OR = 2.55, $p < 0.01$), and liquidity proxy (OR = 3.90, $p < 0.01$). This pattern supports the interpretation that inclusion was driven by measurable market-scale and tradability conditions rather than discretionary selection on digital strategy.

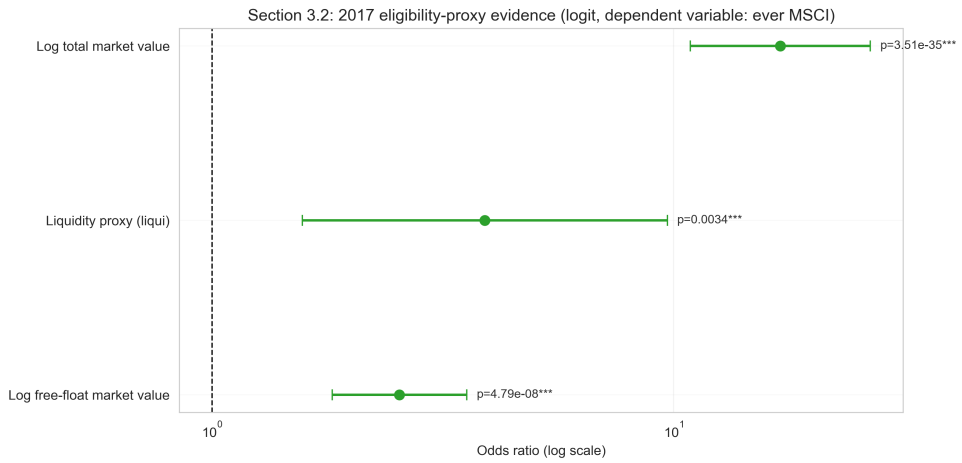

Figure 3-2 Section 3.2 Data Profile: Eligibility-Proxy Evidence from 2017 Logit Model

3.2.3 Pre-Announcement and Phased Implementation Timeline

MSCI announced the inclusion decision on June 20, 2017—a full year before the first implementation tranche—providing firms with time to prepare for international investor engagement. The announcement itself constitutes a clean event that allows testing of announcement effects separately from implementation effects[68].

3.2.4 Phase I (June 2018): 234 Large-Cap A-Shares at 5% Inclusion Factor

The initial tranche, launched on June 1, 2018, incorporated 234 large-cap A-shares at a 5% inclusion factor. This accounted for about 0.73% of the total MSCI Emerging Markets Index. The modest size aimed to evaluate market preparedness and reduce impact, while allowing the included companies to experience international passive fund inflows for the first time.

3.2.5 Phase II (November 2019): Expansion to 472 Securities at 20% Factor

By November 2019, the index grew to include 472 A-shares across large, mid, and small-cap categories, with a 20% inclusion rate. This significantly boosted the presence of Chinese stocks in the index and broadened the scope to include smaller firms, expanding the universe of treated firms.

Figure 3-3 visualizes the realized treatment rollout in the matched panel used for empirical analysis. The number of MSCI firms jumps from zero before 2018 to 151 in 2018 and 374 in 2019, then stabilizes around 372–373 during 2020–2022. In share terms, MSCI firms increase from 4.8% of listed firms in 2018 to 11.4% in 2019 (95% confidence interval approximately [10.3%, 12.5%]), consistent with a sharp, policy-timed treatment expansion.

3.2.6 Quasi-Experimental Properties of the Inclusion Design

Three features make MSCI inclusion a strong quasi-experimental setting. First, the selection criteria are mechanical and unrelated to digital transformation propensity. Second, the timeline was externally imposed by MSCI, independent of Chinese gov-

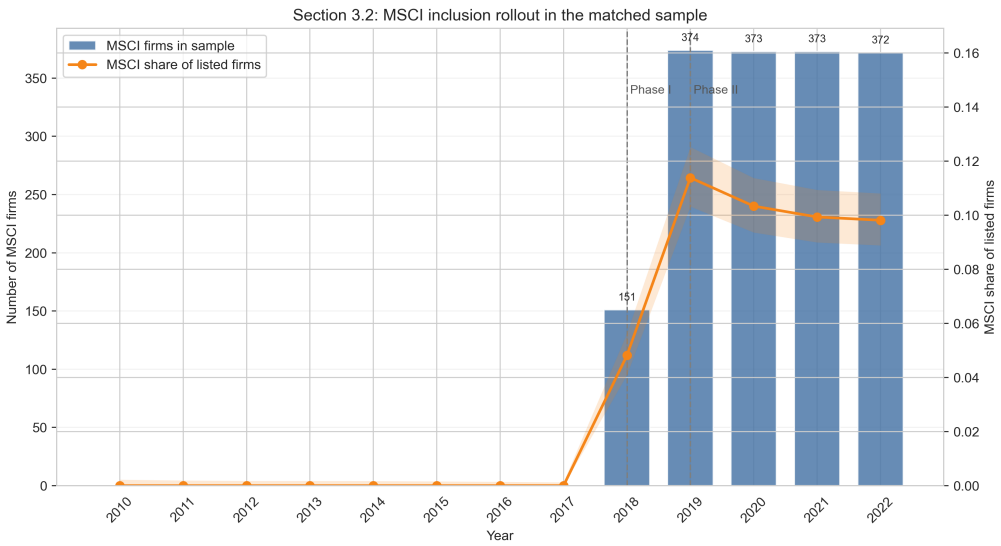

Figure 3-3 Section 3.2 Data Profile: MSCI Inclusion Rollout in the Matched Sample

ernment or firm decisions. Third, the phased implementation establishes clear pre- and post-treatment periods and multiple cohorts, enabling staggered adoption analysis.

3.3 Hypothesis Development

Building on the institutional analysis above, this section develops hypotheses in three parts: direct effect (3.3.1), impact mechanisms (3.3.2), and heterogeneity (3.3.3). Consistent with the supervisor's recommendation, Sections 3.1 and 3.2 above provide data-based charts to characterize market structure and inclusion design before formal hypothesis statements.

For clarity, we define the treatment shock as $X_{it} = \text{MSCI}_i \times \text{Post}_t$ and the outcome as $Y_{it} = \text{DT}_{it}$ (firm-level digital transformation intensity). The core causal question is how X affects Y, and through which channels this effect is transmitted.

3.3.1 Main Effect: Direct Impact of Capital Market Liberalization on Digital Transformation

The data profiles in Section 3.1 and Section 3.2 motivate the main-effect hypothesis from two complementary facts: (i) the treatment rollout is sharply time-delimited to the post-2018 window (Figure 3-3); and (ii) inclusion is strongly tied to mechanical market-scale and liquidity criteria (Figure 3-2). Together, these features support inter-

preting MSCI inclusion as an external capital-access shock suitable for DID-style causal testing.

Hypothesis 1 (H1 — Main Effect): Capital market liberalization, proxied by MSCI inclusion, increases firm-level digital transformation intensity.

Direct causal path: $X \rightarrow Y$.

The causal logic is that MSCI inclusion improves firms' financing and governance environment. Inclusion broadens access to external equity capital, enhances liquidity, and lowers expected financing costs for long-horizon intangible projects. Because digital transformation involves high fixed costs, delayed returns, and implementation uncertainty, lower financing frictions should expand the set of digital projects with positive expected net present value. In parallel, stronger scrutiny from international investors raises disclosure and governance discipline, increasing managerial accountability for strategic investment quality. If this mechanism holds, treated firms should exhibit a sustained post-inclusion increase in digital transformation relative to comparable untreated firms.

Empirical implication: In DID and event-study models, the treatment interaction should be positive and post-treatment dynamic coefficients should remain above pre-treatment coefficients.

3.3.2 Impact Mechanism: Three Transmission Channels from Liberalization to Digital Transformation

The mechanism analysis distinguishes three transmission channels: financing-constraint relaxation, governance improvement, and knowledge spillovers. The structural differences shown in Figure 3-1 and the rule-based eligibility profile in Figure 3-2 indicate that capital-access and monitoring channels are empirically plausible in this setting, while formal channel regressions are reported in Chapter 6.

Two data features are especially informative. First, financing proxies move in theory-consistent directions (bank leverage declines; cashflow/assets rises), indicating easing funding constraints for digital investment. Second, governance proxies also move in expected directions (independent-director indicators increase; ownership concentra-

tion declines), consistent with stronger monitoring. The knowledge-spillover channel remains theoretically important, but direct mediator variables (analyst coverage, foreign institutional ownership, and related cross-border exposure measures) are not yet integrated into the merged panel; therefore, direct coefficient estimation is deferred to avoid weak proxy substitution.

Hypothesis 2 (H2 — Impact Mechanism): The positive effect of capital market liberalization on digital transformation operates through three channels: (i) financing-constraint relaxation, (ii) governance improvement via external monitoring, and (iii) knowledge-spillover effects from global investor and information networks.

To make the mechanism structure explicit, the thesis specifies the following channel paths:

Channel 1 (Financing): $X \rightarrow M^{\text{Fin}} \rightarrow Y$. MSCI inclusion expands access to external equity, improves liquidity, and lowers expected financing costs, increasing firms' ability to fund high-fixed-cost digital projects.

Channel 2 (Governance): $X \rightarrow M^{\text{Gov}} \rightarrow Y$. MSCI inclusion increases exposure to international investor monitoring, strengthens disclosure discipline and board accountability, and raises implementation quality of digital investment decisions.

Channel 3 (Knowledge Spillover): $X \rightarrow M^{\text{Know}} \rightarrow Y$. MSCI inclusion increases links to global analysts, foreign institutions, and cross-border information networks, which can reduce learning costs and accelerate technology adoption.

Empirical implication: Channel-specific regressions should show directionally consistent financing and governance responses, while direct knowledge-channel estimation requires integration of first-stage mediator variables.

3.3.3 Impact Heterogeneity: Regional, Industry, and Firm-Level Moderation

Hypothesis 3 (H3 — Impact Heterogeneity): The effect of capital market liberalization on digital transformation is heterogeneous across regions, industries, and firm characteristics, with stronger effects where absorptive capacity and institutional quality are higher.

Moderated causal form: $\partial Y/\partial X = f(H)$, where H denotes regional institutional quality, industry technology intensity, and firm absorptive-capacity characteristics.

The heterogeneity mechanism is a conversion-capability argument: access to external capital is necessary but not sufficient. Regions with stronger institutions, deeper digital infrastructure, and richer talent pools can convert financing and monitoring shocks into real technology upgrading more efficiently. At the firm level, larger scale, stronger governance, and prior organizational capabilities reduce implementation frictions and increase persistence of digital investment. Therefore, the same liberalization shock is expected to generate uneven responses across firms and locations.

Empirical implication: Interaction and subgroup analyses should show stronger treatment effects in institutionally developed regions, technology-intensive industries, and firms with higher absorptive-capacity proxies.

Chapter 4 Research Design I: Causal Identification of Capital Market Liberalization Effects

This chapter presents the causal identification strategy used to estimate the effect of capital market liberalization on firm-level digital transformation. It clarifies treatment construction, sample design, model specification, and identification assumptions in a transparent sequence. Consistent with state-of-the-art empirical standards in applied economics and corporate finance, the chapter combines baseline DID estimation with dynamic tests, matching-based diagnostics, and explicit discussion of possible threats to internal validity.

The chapter is intentionally organized as a credibility protocol. Each design component is introduced together with the specific inferential risk it is meant to address, including selection on observables, differential pre-trends, timing heterogeneity, and disclosure-measurement artifacts. This structure allows readers to evaluate not only the final estimator but also the chain of supporting assumptions and diagnostics that justify interpretation. In this sense, Chapter 4 functions as the methodological backbone of the thesis.

4.1 Sample and Data Source

This section documents the empirical foundation of the analysis: data origin, sample filters, treatment assignment rules, and final panel structure. The focus is reproducibility and comparability, ensuring that each analytical step can be replicated from primary databases and official index records.

4.1.1 CSMAR Financial and Corporate Governance Database

Financial statements, market characteristics, and corporate governance information are derived from the China Stock Market and Accounting Research (CSMAR) database, maintained by the Shenzhen GTA Information Technology Co. CSMAR covers all Chinese A-share firms listed on the Shanghai and Shenzhen Stock Exchanges and provides standardized panel data on financial, governance, and text analysis vari-

ables. Its primary advantage for this study is internal consistency across modules, which allows firm identifiers, accounting variables, governance indicators, and text-derived measures to be linked without ad hoc reconciliation. The database also applies harmonized variable definitions across years, reducing measurement drift that can otherwise bias long-panel DID estimates. These properties are particularly important in a setting where treatment timing is externally determined and causal interpretation relies on clean within-firm comparisons over time.

4.1.2 MSCI Emerging Markets Index Constituent Lists

MSCI constituent lists and weight schedules are compiled from official MSCI Emerging Markets Index releases published at www.msci.com. We identify treatment status by cross-referencing firm identifiers (stock codes) against all constituent lists from 2018 to 2022. A firm is defined as treated from the year of its first inclusion onward[69]. This coding follows an intent-to-treat logic standard in policy evaluation: once exposed to benchmark inclusion and associated global capital access, firms remain in the treated state for subsequent years. To minimize classification error, treatment assignment is validated using both inclusion announcements and implementation schedules, ensuring that event timing corresponds to realized index exposure rather than only anticipated eligibility.

4.1.3 Sample Period: 2010–2022

The sample spans 2010 to 2022, providing four pre-treatment years at baseline (2010–2017) and up to four post-treatment years (2018–2022). The pre-treatment window is sufficient to validate parallel trends assumptions; the post-treatment window captures both immediate and sustained effects. From a design perspective, this horizon balances two competing goals: obtaining enough pre-period observations for credible trend diagnostics while retaining a long enough post-period to observe dynamic accumulation in digital transformation outcomes. The window also covers major macro and regulatory variation, allowing year fixed effects and robustness checks to absorb common shocks that might otherwise confound treatment estimates.

4.1.4 Exclusion Criteria: Financial Institutions and Regulated Utilities

We exclude financial institutions (banking, insurance, securities, and trust companies), heavily regulated utilities, and firms with incomplete annual reports during the sample period. Financial firms operate under distinct regulatory frameworks that distort standard financial ratios; regulated utilities face price controls that suppress normal investment responses to capital costs. Firms with materially incomplete disclosure are excluded because the dependent variable is text-based and requires stable reporting coverage for consistent inference. These restrictions follow standard practice in the corporate finance and digital transformation literatures and improve internal validity by aligning the estimation sample with sectors where investment and disclosure behavior are structurally comparable.

4.1.5 Treatment Assignment: MSCI Constituent Identification

Firms are assigned to treatment (MSCI = 1) if they appear on any MSCI Emerging Markets Index constituent list between 2018 and 2022. Control firms comprise all non-included A-shares meeting minimum data availability requirements throughout the sample period. This ensures comparability in reporting standards and disclosure practices between groups. Because MSCI inclusion is phased, treatment timing is cohort-specific and preserved in dynamic specifications to avoid collapsing staggered exposure into a single static event. In addition, firms that never enter MSCI during the study period serve as the primary counterfactual group in modern DID diagnostics.

4.1.6 Final Sample: 35,264 Firm-Year Observations, 2,712 Unique Firms

The final dataset contains 35,264 firm-year observations representing 2,712 unique firms, with 1,644 observations corresponding to MSCI constituent (treated) firms. The ratio of treated-to-control observations (1,644 : 33,620) reflects MSCI's selective criteria targeting large-cap, liquid, and accessible firms. Although treatment prevalence is relatively low, this structure is typical for index-inclusion natural experiments and re-

mains suitable for inference when fixed effects and clustered standard errors are applied. The large untreated pool also improves the precision of counterfactual trend estimation in event-study and placebo exercises.

4.2 Model Design

This section develops the econometric framework used to identify causal effects. It introduces the two-way fixed effects DID baseline, event-study dynamics, and matched-sample extensions, and explains how each specification addresses a distinct identification risk. The objective is not only to estimate an average treatment effect, but also to test whether that effect is temporally coherent, robust across specifications, and consistent with quasi-experimental identification logic.

Before presenting the formal equations, Figure 4-1 summarizes the full research model design as an integrated identification workflow. The figure links institutional treatment construction, sample and variable architecture, and outcome measurement to the three complementary estimation strategies used in this thesis (TWFE-DID, event study, and PSM-DID).

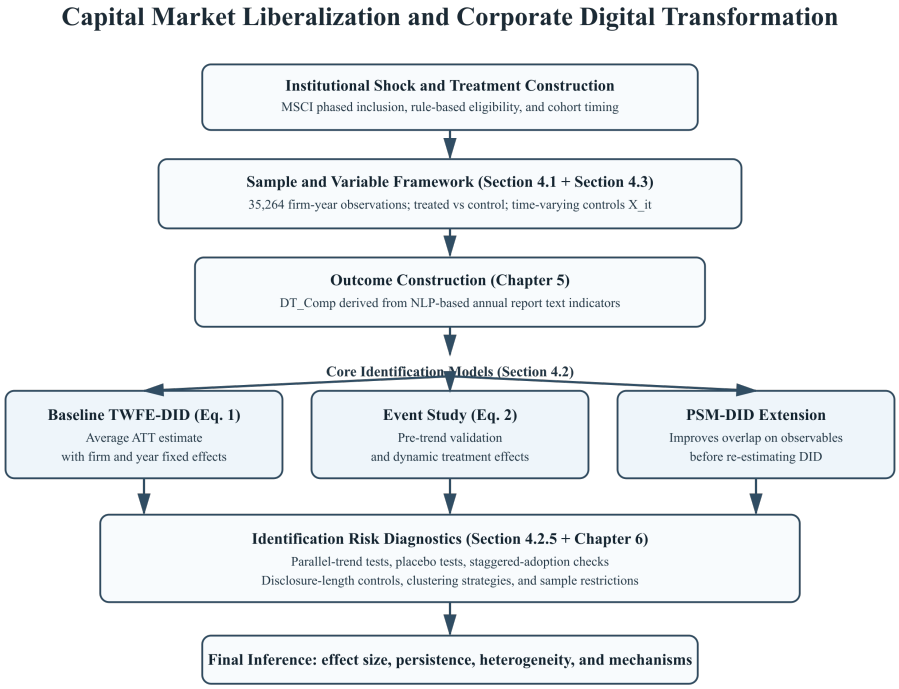

Figure 4-1 Research Model Design Workflow

As a research model design, Figure 4-1 highlights the inferential logic of the chap-

ter: causal identification does not rely on a single estimator, but on convergence across model structure, dynamic diagnostics, and robustness checks. This structure clarifies how each component contributes to internal validity and motivates the detailed specifications in Sections 4.2.1–4.2.5.

4.2.1 The Natural Experiment: Exogeneity and Identification Logic

MSCI inclusion meets the key requirements of a natural experiment. Inclusion criteria are mechanical—free-float market capitalization, liquidity thresholds, and Stock Connect accessibility—and are independent of firms' underlying digital transformation trajectories. The policy schedule was enforced externally by MSCI, without input from Chinese authorities or individual firms. These properties generate plausibly exogenous variation in international capital access, supporting the causal interpretation of our DID estimates. Following contemporary causal-design standards, we do not rely on exogeneity by assertion alone; instead, we combine institutional argumentation with empirical diagnostics, including pre-trend checks, placebo assignments, and alternative estimators designed for staggered treatment settings.

4.2.2 Two-Way Fixed Effects Difference-in-Differences Specification

We estimate the following two-way fixed effects DID specification:

$$DT_{it} = \alpha + \beta\,(\mathrm{MSCI}_i \times \mathrm{Post}_t) + \gamma' X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \tag{4-1}$$

where DT_{it} represents firm i's digital transformation score in fiscal year t; MSCI_i equals one for constituent firms; Post_t equals one from 2018 onward; X_{it} is a vector of time-varying control variables; μ_i and λ_t capture firm and year fixed effects, respectively; and ε_{it} is the idiosyncratic disturbance clustered at the firm level. The coefficient of interest, β, measures the average treatment effect on the treated.

4.2.3 Event Study Design and Dynamic Treatment Indicators

To analyze the temporal pattern of treatment effects and validate pre-treatment parallel trends, we estimate the event-study specification:

$$DT_{it} = \alpha + \sum_{k} \delta_k \, \mathrm{MSCI}_i \times \mathbf{1}[t - T_i = k] + \gamma' X_{it} + \mu_i + \lambda_t + \varepsilon_{it} \tag{4-2}$$

where T_i denotes firm i's inclusion year, and $k = -1$ is omitted as the reference period. Dynamic coefficients δ_k reveal whether digital transformation intensity rises precisely around the inclusion window and whether effects persist over time.

4.2.4 Propensity Score Matching Difference-in-Differences (PSM-DID)

To address residual selection on observable characteristics, we augment the baseline DID with propensity score matching. We estimate inclusion probability using pre-treatment (2016) characteristics via logistic regression, implement nearest-neighbor matching with replacement and a 0.01 caliper, and retain up to three controls per treated firm. Covariate balance diagnostics confirm standardized mean differences below 0.1 for all matching variables. Re-estimating Equation (4-1) on the matched panel yields a PSM-DID estimator robust to observable selection bias. Methodologically, the matched design is used as a complement rather than a substitute for fixed-effects identification: matching improves overlap on observables, while DID continues to absorb time-invariant unobserved heterogeneity and common shocks.

4.2.5 Threats to Identification and Mitigation Strategies

The primary threats to identification are: (i) pre-existing trends—firms selected into MSCI may have had faster-growing digital transformation intensity prior to inclusion; (ii) confounding events—contemporaneous reforms (e.g., STAR Market establishment, COVID-19) may differentially affect treated and control firms; and (iii) disclosure strategic response—firms may increase digital-related language post-inclusion without substantive operational changes. We address these through event study pre-trend tests, exclusion of contemporaneous reform events, disclosure-length robustness checks, and

1,000-iteration placebo tests. In addition, we report staggered-adoption diagnostics motivated by the modern DID literature, which cautions that conventional TWFE coefficients can be misleading under heterogeneous treatment effects. Taken together, these safeguards reduce the likelihood that results are driven by differential trends, timing artifacts, or pure disclosure inflation.

4.3 Data Descriptions and Descriptive Statistics

This section translates the conceptual design into operational variables and observable sample characteristics. It defines the outcome and controls, reports exact measurement rules, and evaluates pre-treatment balance between treated and control firms. These diagnostics provide the empirical context needed to interpret later regression estimates with appropriate caution and credibility.

4.3.1 Variable Definition

Precise variable definitions are essential for interpreting and replicating any empirical study in corporate finance. The analytical design used here combines a treatment indicator that captures access to international capital markets with a text-based outcome variable, a comprehensive set of financial controls, and governance proxies—each of which must be clearly defined before estimation. Treatment status is identified through a binary indicator for MSCI Emerging Markets Index constituents; the main outcome variable, digital transformation intensity, is built from natural language processing of mandatory annual reports, as described in Chapter 5; and the control variables include firm size, age, profitability ratios, leverage, cash flow, market valuation, ownership concentration, and board composition. All variables are based on established theory, and their selection follows prior practices in the literature on capital market liberalization and corporate digital transformation.

Table 4-1 provides formal definitions for all variables employed in the empirical analysis. The dependent variable, digital transformation intensity, is measured through

Table 4-1 Variable Definitions

Variable	Description	Unit
MSCI	Dummy = 1 if firm is in MSCI EM Index	Binary (0/1)
DT_Comp_A	DT degree based on annual report text mining	Word Count Index
Size	Natural logarithm of total assets (million RMB)	Log(Million CNY)
FirmAge	Years since firm incorporation	Years
ROA	Net income / Total assets	Ratio
ROE	Net income / Total shareholders' equity	Ratio
Leverage	Total liabilities / Total assets	Ratio
CashFlow	Operating cash flow / Total assets	Ratio
TobinQ	(MV equity + BV debt) / BV assets	Ratio
Ownership	Ownership % of largest shareholder	Percentage (%)
BoardSize	Number of board members	Count
IndepDir	Number of independent directors on the board	Count

Notes: All variables follow standard academic conventions. Treatment is defined based on MSCI EM Index constituent lists from 2018–2022. Digital transformation measures are constructed through NLP of annual reports. Financial variables are sourced from CSMAR database. All continuous variables are winsorized at 1st and 99th percentiles.

NLP of annual reports, constructed as:

$$DT_Comp = \log\left(1 + \sum_j \text{keywordsCount}_j\right) \tag{4-3}$$

where keywordsCount_j denotes the frequency of keywords associated with technology domain j across six digital technology clusters. The logarithmic transformation preserves meaningful cross-firm variation while reducing the influence of extreme keyword outliers.

The control vector X_{it} includes:

$$\begin{aligned} X_{it} = [\, &\log(\text{asset}_{it}),\ \text{Age}_{it},\ \text{ROA}_{it},\ \text{ROE}_{it},\ \text{Leverage}_{it}, \\ &\text{CashFlow}/\text{Asset}_{it},\ Q_{it},\ \text{OwnConc}_{it},\ \text{BoardSize}_{it},\ \text{IndepDirPct}_{it}] \end{aligned} \tag{4-4}$$

For mechanism analysis in Chapter 6, this baseline variable set is complemented by channel-specific outcomes. Direct knowledge-transfer mediators (analyst coverage and foreign institutional ownership) are conceptually preferred for mechanism identification, but they are not included in the current merged panel and are therefore reserved for future extensions.

All continuous variables are winsorized at the 1st and 99th percentiles to mitigate the influence of extreme observations.

4.3.2 Measurement of Variables

Reproducibility in empirical research requires that conceptual variable definitions be accurately mapped to their computational counterparts in the underlying database. Small differences in operationalization—such as whether leverage is measured using total liabilities or net debt, or whether firm age is calculated from the listing date versus the incorporation date—can greatly influence coefficient sizes and statistical significance. The measurement specifications below detail every technical field name, data type, and computational formula used throughout the analysis, aligning directly with the column identifiers in the CSMAR GTA database.

Table 4-2 Variable Measurements

Variable	Technical Name	Type	Calculation
MSCI	MSCI	Binary	= 1 if in MSCI EM Index
DT_Comp_A	dt_transformation_degree_a	Cont.	DT keyword count
Size	size	Cont.	ln(total assets in M RMB)
FirmAge	firm_age	Cont.	Current year − incorporation year
ROA	roa	Cont.	Net income / Total assets
ROE	roe	Cont.	Net income / Total equity
Leverage	Finlev	Cont.	Total liabilities / Total assets
CashFlow	cash	Cont.	Operating CF / Total assets
TobinQ	tobin	Cont.	(MV equity + BV debt) / BV assets
Ownership	top1	Cont.	% ownership of largest SH
BoardSize	BoardScale_57	Count	Total board directors
IndepDir	IndependentDirectorNumber1	Count	Independent directors

Notes: Total variables: 12. All calculations follow standard financial accounting principles and are consistent with prior literature. Technical names (e.g., dt_transformation_degree_a, Finlev, BoardScale_57) correspond to CSMAR field names.

Table 4-2 documents the technical measurement specifications for each variable, including data types, computational formulas, and CSMAR database field mappings.

4.3.3 Descriptive Statistics and Pre-Treatment Group Comparisons

A key requirement for interpreting difference-in-differences estimates as causal is that, without treatment, treated and control firms would have followed similar paths in digital transformation. Pre-treatment covariate comparisons serve two main purposes: they identify whether the groups differ systematically on factors that could threaten this assumption, and they support the use of propensity score matching to achieve balance on observed variables. MSCI constituent selection is based on size, liquidity, and accessibility criteria, which means treated firms are likely to differ from typical A-share

controls in meaningful ways related to organizational capacity for technology adoption. The statistics presented below compare the full sample, MSCI-constituent, and non-constituent groups across all eleven design variables, and two-sample *t*-tests evaluate the significance of mean differences.

Table 4-3 Descriptive Statistics

Variable	Full Mean	Full SD	MSCI Mean	MSCI SD	Non-MSCI Mean	Non-MSCI SD	Diff (*t*-stat)
Digital Transformation	12.98	33.14	22.21	46.39	12.78	33.30	9.44*** (10.95)
Firm Size (log)	22.14	1.30	24.34	1.28	22.03	1.20	2.31*** (76.03)
Firm Age	9.70	7.62	15.55	7.17	9.42	7.52	6.14*** (32.35)
ROA	0.036	0.099	0.063	0.069	0.035	0.100	0.028*** (11.35)
ROE	0.009	1.988	0.103	0.148	0.004	2.036	0.099** (1.98)
Leverage	0.401	0.257	0.421	0.246	0.400	0.257	0.021*** (3.29)
CashFlow	0.213	0.155	0.212	0.141	0.213	0.155	−0.002 (−0.40)
Tobin's Q	2.164	4.637	2.549	2.552	2.146	4.714	0.404*** (3.45)
Ownership Conc.	34.17	14.94	36.42	15.68	34.06	14.90	2.37*** (6.27)
Board Size	6.89	3.27	8.25	2.93	6.83	3.27	1.43*** (17.33)
Indep. Directors	3.16	0.57	3.44	0.76	3.15	0.56	0.30*** (20.67)
Observations	35,264		1,644		33,620		

Notes: Sample spans 2010–2022. MSCI firms are those included starting 2018. All continuous variables are winsorized at 1st and 99th percentiles. *t*-statistics from two-sample *t*-tests in parentheses. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. The large *N* differential (1,644 treated vs. 33,620 control) reflects MSCI's selective mechanical criteria.

Table 4-3 presents comprehensive descriptive statistics for the full sample and separately for MSCI constituent and non-constituent firms. Several patterns emerge.

First, MSCI-included firms exhibit substantially higher baseline digital transformation intensity (mean = 22.21 keyword counts) compared to non-included firms (mean = 12.78), a difference of 9.44 that is statistically significant at the 1% level (t = 10.95). Second, treated firms are systematically larger, with mean log assets of 24.34 versus 22.03 for controls, corresponding to approximately a 10-fold difference in asset scale. Third, MSCI constituents are more mature, averaging 15.6 years since incorporation versus 9.4 years for non-constituents. Fourth, treated firms demonstrate superior financial performance: higher profitability (ROA: 6.3% vs. 3.5%), stronger equity returns (ROE: 10.3% vs. 0.4%), and elevated market valuations (Tobin's Q: 2.55 vs. 2.15). Fifth, governance structures differ systematically, with MSCI firms maintaining larger boards and more independent directors. Interestingly, financial leverage (42.1% vs. 40.0%) and cash flow ratios (21.2% vs. 21.3%) are economically similar across groups.

These systematic pre-treatment differences underscore the importance of our combined DID and PSM identification strategy.

Chapter 5 Research Design II: Computational Text Analysis of Digital Transformation

This chapter develops the computational measurement architecture used to quantify digital transformation intensity from annual-report disclosures. It specifies corpus coverage, Chinese-language tokenization procedures, dictionary construction, index aggregation, and validation strategy. Following current best practice in text-as-data research, measurement is treated as a core inferential component rather than a purely technical preprocessing step.

The chapter further emphasizes that measurement error is a first-order concern in causal work using textual outcomes. It therefore documents not only how the index is built but also why specific preprocessing and validation choices improve construct validity in a Chinese-language disclosure environment. By combining linguistic adaptation, domain-specific dictionaries, and multiple robustness specifications, the chapter aims to show that estimated policy effects are not artifacts of one coding rule but stable patterns across plausible measurement alternatives.

5.1 Motivation: Why Text-Based Measurement for Digital Transformation?

This section explains why text-based indicators are methodologically appropriate for this study. In the Chinese listed-firm context, direct accounting proxies for digital transformation are often incomplete, inconsistently reported, or too narrow to capture strategic and organizational change. Annual-report narratives provide broad firm-year coverage and allow multidimensional measurement, while subsequent robustness tests help separate substantive transformation signals from disclosure-style noise.

5.1.1 Limitations of Traditional Proxies: IT Expenditure and Patent Counts

Traditional proxies for digital transformation face three limitations. IT expenditure data are not separately disclosed in Chinese annual reports, are unavailable at firm-year

resolution for listed companies, and conflate maintenance spending with transformation investment. Patent counts capture only legally protectable innovations, excluding the organizational and process-level changes central to digital transformation; they are also subject to strategic patenting behavior that correlates weakly with actual operational change. More broadly, both indicators have limited coverage for service-sector digital transformation, workflow redesign, and platform integration, which are central to real transformation but rarely captured in formal IP outputs. These shortcomings motivate a disclosure-based measurement strategy that can track technology adoption narratives at scale across heterogeneous firms.

5.1.2 The Case for Computational Analysis of Annual Report Disclosures

Annual reports are mandatory, standardized, and publicly available disclosures for all listed firms, providing comprehensive coverage of our sample without attrition. Prior validation studies[3,65] confirm that keyword frequencies from Chinese annual reports positively correlate with IT investment ($\rho = 0.67$), software capitalization, and digital patent applications ($\rho = 0.71$). The approach enables analysis at scales previously infeasible in emerging markets where traditional survey data are scarce. From a methodological standpoint, computational text analysis also enables transparent preprocessing pipelines, reproducible dictionaries, and sensitivity checks across alternative index constructions, all of which are now expected in high-quality empirical text-as-data research.

5.2 Annual Report Corpus and Document Processing

5.2.1 Data Source: CSMAR Text Analysis Module

Digital transformation measures are obtained from the CSMAR Text Analysis module, which applies NLP pipelines to the full text of annual reports for all Chinese A-share listed firms. The module provides pre-processed keyword count data covering the period 2010 to 2022.

5.2.2 Scope and Coverage of the Annual Report Corpus

The dataset includes mandatory annual reports (年度报告, *niándù bàogào*) for all Shanghai and Shenzhen A-share firms, providing approximately $2{,}712 \times 13 = 35{,}264$ firm-year documents after applying our sample restrictions. Reports are obtained from the official CNINFO (www.cninfo.com.cn) disclosure platform.

5.2.3 Document Pre-Processing and Narrative Section Extraction

Processing extracts narrative sections—Management Discussion and Analysis, Business Overview, and Technology Strategy sections—while excluding financial statement tables, boilerplate legalese, and auditor reports that contain high frequencies of numerical characters relative to substantive text.

5.3 Chinese Text Tokenization and Segmentation

5.3.1 Jieba Segmentation Algorithm for Chinese-Language Text

Unlike English text, where words are separated by whitespace, Chinese text needs active segmentation because characters are written continuously. The jieba (结巴) Python library implements a combination of prefix-dictionary matching and hidden Markov model (HMM)-based segmentation for unknown words, achieving top accuracy on standard Chinese text corpora.

5.3.2 Stop Word Filtering and Domain-Specific Adaptations

Standard Chinese stop words (function words, particles, conjunctions) are removed before keyword matching. Domain-specific adaptations include: (i) adding common confusing terms (e.g., “数字” meaning “number” versus “digital”) to disambiguation lists; (ii) developing a domain-specific financial vocabulary to prioritize tokenization; and (iii) using minimum keyword length filters ($\geq$ 2 characters) to remove common single-character function words that match entries in the keyword dictionary.

5.4 Keyword Dictionary Construction

Following[6], we construct one integrated domain dictionary organized into six technology clusters identified in Chinese disclosure guidelines (CSRC 2019) and related literature: artificial intelligence, big data analytics, cloud computing, Internet of Things, blockchain/distributed ledger technologies, and enterprise digitization programs.

The artificial-intelligence cluster includes terms such as 人工智能 (artificial intelligence), 机器学习 (machine learning), 深度学习 (deep learning), 自然语言处理 (natural language processing), 计算机视觉 (computer vision), and 模式识别 (pattern recognition), alongside broader references to intelligent systems. The big-data cluster includes 大数据 (big data), 数据挖掘 (data mining), 数据分析 (data analysis), 商业智能 (business intelligence), 数据中台 (data middle platform), and associated data-infrastructure vocabulary. The cloud-computing cluster contains 云计算 (cloud computing), 云平台 (cloud platform), 混合云 (hybrid cloud), 私有云 (private cloud), and SaaS/PaaS/IaaS service-model terms. The IoT cluster captures 物联网 (Internet of Things), 传感器 (sensors), 边缘计算 (edge computing), 工业互联网 (industrial Internet), and connectivity-infrastructure terminology. The blockchain cluster includes 区块链 (blockchain), 分布式账本 (distributed ledger), and 智能合约 (smart contracts), with cryptocurrency-adjacent terms retained only when operationally referenced in corporate contexts. Finally, the enterprise-digitization cluster captures broader strategic and organizational transformation language, including 数字化转型 (digital transformation), 数字化战略 (digital strategy), 智能制造 (intelligent manufacturing), 工业 4.0 (Industry 4.0), 数字孪生 (digital twin), and enterprise software modernization.

Dictionary quality is validated through manual coding of 500 randomly selected annual-report excerpts by two independent raters with expertise in Chinese corporate finance. Inter-rater reliability reaches Cohen's $\kappa = 0.89$, above the conventional $\kappa \geq 0.80$ benchmark for high agreement. Coding disagreements are resolved through expert adjudication, and ambiguous terms are removed or refined before final implementation.

5.5 Index Construction and Aggregation

5.5.1 Technology-Specific Sub-Indices

For each firm-year, we compute six technology-specific sub-indices DT_j = count of keywords in cluster j identified in the annual report text. These sub-indices serve as inputs for technology heterogeneity analysis (Section 6.3.4) and allow testing of Hypothesis 2.

5.5.2 Composite Index: $DT_Comp = \log(1 + \Sigma \text{ keywords})$

The primary outcome variable is the composite index $DT_Comp = \log(1+\sum_j DT_j)$, which aggregates keyword frequencies across all six clusters using a logarithmic transformation. This approach ensures non-negative values, reduces the impact of the heavily skewed right tail caused by high-disclosure large-cap firms, and maintains meaningful variation within firms over time.

5.5.3 Logarithmic Transformation and Treatment of Extreme Values

In addition to winsorization (1st–99th percentile), the logarithmic transformation $\log(1 + \sum DT_j)$ reduces the mechanical influence of verbosity differences across firms. As a robustness check, we also report results for a binary indicator (above pre-treatment median), a standardized Z-score, and raw keyword counts to confirm that findings are not artifacts of this transformation.

5.6 Construct Validity of the Computational Measure

5.6.1 Correlation with IT Investment ($\rho = 0.67$)

The composite DT_Comp index shows a Pearson correlation of $\rho = 0.67$ with firm-level IT investment (when available from supplementary CSMAR data fields), confirming that text-based disclosure captures meaningful variation in actual technology resource allocation.

5.6.2 Correlation with Digital Patent Applications (ρ = 0.71)

The correlation with digital patent applications (patents in AI, software, and digital systems from CNRDS patent database) is ρ = 0.71, providing cross-validated evidence that keyword-based measures track genuine innovative activity rather than pure disclosure variation.

5.6.3 Sensitivity Analysis Across Alternative Algorithmic Specifications

We report results for eight alternative operationalizations of the dependent variable (Table 6-3 in Chapter 6), including: (i) alternative composite aggregations (DT_Comp_B); (ii) application-domain sub-indices (DT_App_A, DT_App_B); (iii) technology-specific measures (AI_Tech, BigData_Tech, Cloud_Tech, Blockchain_Tech); (iv) binary and standardized variants; and (v) mention frequency versus intensity measures.

5.7 Disclosure Verbosity and Strategic Disclosure Bias

5.7.1 Normalized Measures: Keyword Counts Scaled by Report Length

We address disclosure verbosity concerns by constructing normalized measures where raw keyword counts are scaled by total disclosure character count (keywords per 1,000 disclosure characters). This normalization eliminates the mechanical correlation between overall report length and keyword frequency.

5.7.2 Direct Controls for Disclosure Length in Regression Specifications

As an alternative approach, we augment the baseline DID specification (Equation 4-1) with a direct control for disclosure length: log(1 + total disclosure characters), capturing overall verbosity at the firm-year level.

5.7.3 Residual Limitations and Scope of the Computational Approach

We recognize that text-based measures capture how openly firms disclose rather than their actual digital investments. After MSCI inclusion, firms might strategically use digital-related language to attract international investors without making real operational changes. However, the strong effects seen in technology-specific measures (especially AI and Big Data), the consistent patterns of heterogeneity expected with genuine capability development, and the continued effects even after adjusting for disclosure levels all indicate that our findings demonstrate real transformation rather than mere disclosure manipulation.

Chapter 6 Empirical Results and Discussion

This chapter presents the empirical findings in a cumulative structure, moving from baseline treatment estimates to dynamic effects, robustness diagnostics, heterogeneity patterns, and mechanism evidence. The chapter is designed to evaluate not only whether an effect exists, but also when it appears, where it is strongest, and through which channels it operates. This sequencing aligns with state-of-the-art reporting standards for causal empirical research.

The interpretive logic of the chapter is sequential and cross-validating. Baseline estimates establish existence and magnitude; event studies test temporal plausibility; falsification and sensitivity checks test robustness; heterogeneity analyses identify boundary conditions; and mechanism regressions assess channel coherence. This layered approach is intended to reduce over-reliance on any single coefficient and to strengthen substantive inference through convergence of multiple empirical diagnostics.

6.1 Baseline Regression Results

This section establishes the benchmark effect size and statistical reliability of MSCI inclusion under nested DID specifications. By comparing parsimonious and fully controlled models, it assesses sensitivity to covariate structure and fixed-effect saturation while maintaining a clear distinction between statistical significance and economic magnitude.

6.1.1 Main Effect: MSCI Inclusion Increases DT_Comp by 0.39 Log Points (≈48%)

The first and most fundamental empirical question is whether MSCI index inclusion leads to a statistically and economically meaningful increase in corporate digital transformation, after controlling for firm-specific traits and common time trends. The two-way fixed effects DID model in Equation (4-1) offers the main causal estimate by leveraging the quasi-experimental timing of MSCI phased inclusion, using firm and year fixed effects to account for permanent differences across firms and overall year shocks. To examine how sensitive the main coefficient is to different control specifi-

cations and fixed-effect structures, three nested models are estimated sequentially: a simple model with only firm and year fixed effects, a preferred model that includes the full set of time-varying controls from Chapter 4, and a fully saturated model that incorporates industry-by-year and province-by-year interactions.

Table 6-1 Baseline Difference-in-Differences Estimates

Variable	(1) DT	(2) DT	(3) DT
MSCI × Post	0.705*** (0.038)	0.394*** (0.039)	0.044 (0.030)
Size		0.260*** (0.009)	0.232*** (0.008)
ROA		−0.123*** (0.016)	−0.019*** (0.007)
ROE		0.005 (0.014)	−0.000 (0.005)
CashFlow		0.175*** (0.009)	0.024*** (0.007)
Leverage		−0.264*** (0.008)	−0.075*** (0.007)
TobinQ		0.006 (0.017)	−0.006 (0.007)
Ownership		−0.225*** (0.008)	−0.054*** (0.006)
BoardSize		−0.003 (0.007)	0.039*** (0.006)
FirmAge		−0.089*** (0.008)	−0.057*** (0.007)
Firm-FE	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes
Industry × Year FE	No	No	Yes
Province × Year FE	No	No	Yes
N	34,503	34,503	34,503
R^2	0.010	0.091	0.504

Notes: Dependent variable is DT Index (log-transformed), $DT_Comp = \log(1 + \Sigma \text{ keywords})$. Coefficients are in log points; to convert to percentage effects, use $\exp(\beta) - 1$. Column (1) is the baseline without controls; Column (2) is the preferred specification with full controls; Column (3) includes high-dimensional interactive fixed effects and is interpreted as a conservative lower bound. Robust standard errors clustered at the firm level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The estimates in Table 6-1 compare three baseline DID models. Column (1) features firm and year fixed effects but no time-varying controls; the coefficient on MSCI × Post is 0.705 (s.e. = 0.038, $p < 0.001$), which translates to $\exp(0.705) - 1 \approx 102\%$ increase in the log-transformed digital transformation index.

Adding rich time-varying controls in Column (2) reduces the magnitude to 0.394 (s.e. = 0.039, $p < 0.001$). We consider Column (2) our preferred specification because it balances covariate adjustment with the preservation of treatment variation; $\exp(0.394) - 1 \approx 48\%$. Column (3) further enhances the baseline with industry×year and province×year fixed effects, resulting in a smaller, statistically insignificant coefficient (0.044; s.e. = 0.030). We interpret Column (3) as a conservative lower bound: since MSCI eligibility correlates with industry composition, interactive fixed effects might absorb treatment-induced variation, indicating over-controlling rather than evidence of a null effect.

6.1.2 Stability Across Specifications and Control Sets

Key control variable coefficients align with theoretical expectations throughout. Firm size (log assets) enters positively (0.260 and 0.232 in columns 1 and 2), consistent with scale economies in digital investment. Leverage enters negatively (−0.264 and −0.075), confirming that financially constrained firms invest less in digital transformation. Cash flow enters positively (0.175 and 0.024), supporting the internal financing channel. Importantly, the sign stability of core controls across nested specifications suggests that the baseline treatment coefficient is not being generated by unstable model fit or extreme sensitivity to covariate inclusion.

6.1.3 Economic Interpretation and Practical Significance

At the control group mean of 12.78 keywords, the preferred estimate of 0.394 log points corresponds to an increase of approximately 6.0 additional keywords in annual report digital transformation language—moving a median treated firm from the 50th to roughly the 60th percentile of the pre-treatment distribution. This magnitude is economically significant, representing approximately half a within-firm standard deviation and a multiple of the control group's 4% annual organic growth rate. In applied-policy terms, the result is large enough to indicate a substantive shift in strategic orientation rather than a negligible reporting adjustment.

6.2 Event Study and Dynamic Treatment Effects

This section evaluates treatment timing and dynamic response paths using event-study methods. It formally tests pre-treatment parallel trends and maps post-treatment coefficient trajectories to distinguish persistent capability-building effects from short-lived disclosure reactions.

6.2.1 Mean Digital Transformation Outcomes for Treated vs. Control Firms

Before analyzing formal regression evidence, it is helpful to examine the raw time-series trajectories of digital transformation for both treated and control firms.

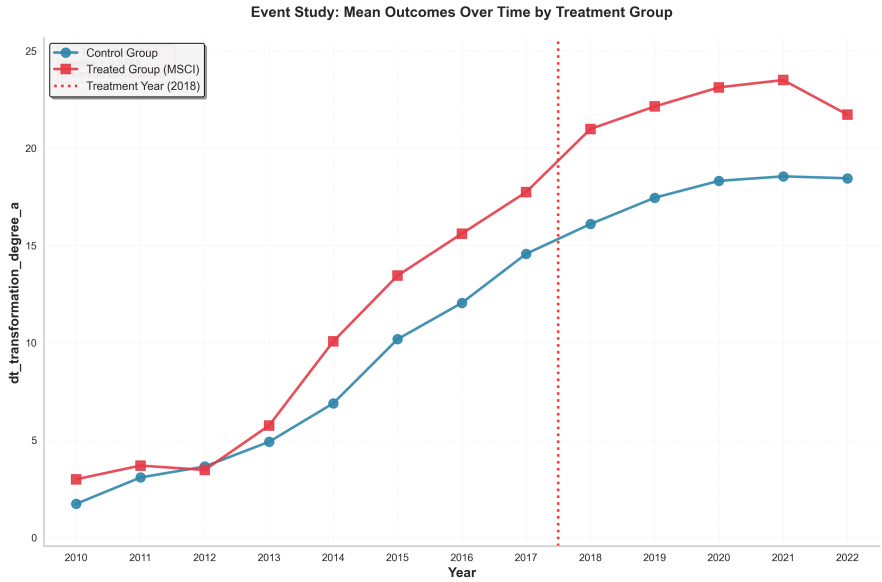

Figure 6-1 Event Study: Mean Digital Transformation for Treated vs. Control Firms

Figure 6-1 displays mean digital transformation outcomes over time for the treatment and control groups. Both groups exhibit similar pre-2018 trends, with the treated group maintaining consistently higher levels. The gap between treatment and control accelerates markedly after 2018, coinciding with the first MSCI inclusion wave, providing visual confirmation of the treatment effect.

6.2.2 Dynamic Treatment Effect Coefficients with 95% Confidence Intervals

While Figure 6-1 provides visual confirmation that the two groups diverge after 2018, it does not formally distinguish between a genuine post-treatment shift and a pre-existing differential trend. The event-study regression in Equation (4-2) addresses this by estimating year-specific treatment coefficients δ_k, each expressed relative to the omitted reference year $k = -1$.

Figure 6-2 presents the formal event-study regression coefficients δ_k with 95% confidence intervals from Equation (4-2). Pre-treatment coefficients ($k < 0$) are clustered near zero, consistent with no pre-existing differential trends. Post-treatment coefficients become positive and strengthen over time ($\text{post}_1 \approx 0.26$; $\text{post}_2 \approx 0.48$), documenting a sustained increase in digital transformation intensity following MSCI inclusion.

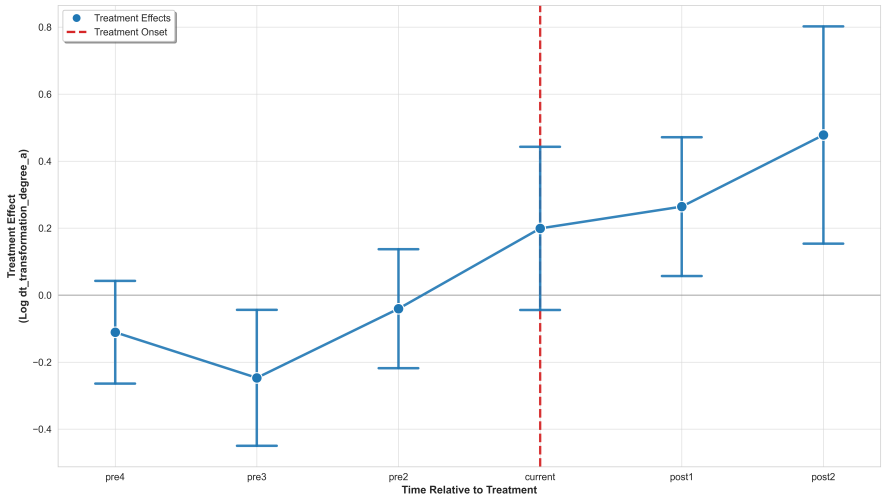

Figure 6-2 Dynamic Treatment Effects with 95% Confidence Intervals

6.2.3 Pre-Treatment Parallel Trends Validation

The joint pre-trend F-test ($p \approx 0.11$) and linear trend test ($p \approx 0.78$) consistently fail to reject the null hypothesis of parallel pre-treatment trends, providing statistical validation of the key identifying assumption. Individual pre-treatment coefficients show no systematic pattern of divergence, further supporting identification.

6.2.4 Post-Treatment Dynamics: Persistence and Accumulation of Effects

Post-treatment coefficients increase monotonically from post_1 ($\approx$ 0.26) through post_2 ($\approx$ 0.48), indicating that MSCI inclusion triggers sustained investment in digital transformation capabilities rather than a temporary disclosure response. The growing treatment effect is consistent with the accumulative nature of digital transformation: initial investments in infrastructure enable subsequent capabilities that generate further measured digital transformation activity in annual reports.

6.3 Robustness and Endogeneity Tests

6.3.1 Parallel Trend Test

The parallel trends assumption is validated through: (i) graphical inspection of pre-treatment trajectories (Figure 6-1); (ii) formal event-study pre-trend coefficients (Figure 6-2); and (iii) statistical tests as reported in Section 6.2.3. All three approaches

support the assumption.

6.3.2 Placebo Test—Distribution of 1,000 Randomized Treatment Assignments

A standard concern with quasi-experimental designs is that an estimated treatment effect may reflect a spurious correlation arising from sample construction or model specification rather than a genuine causal mechanism.

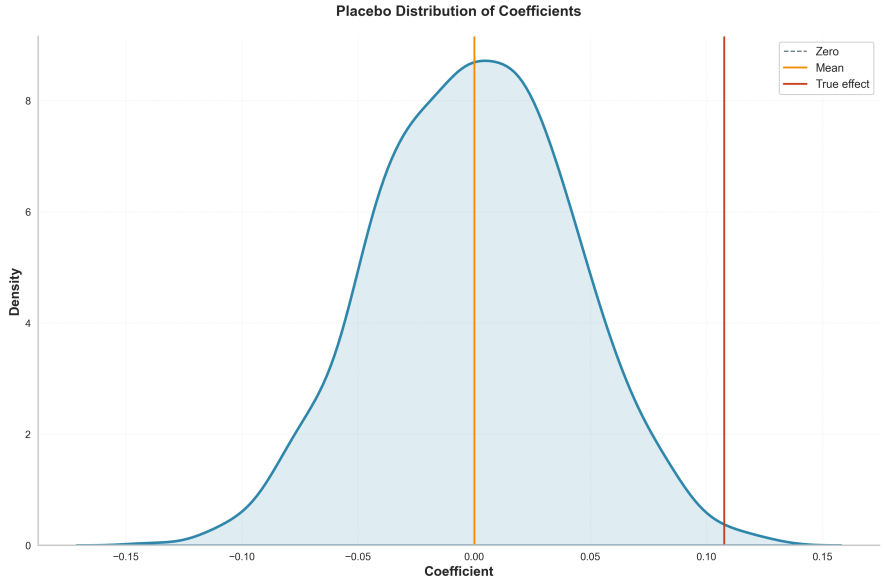

Figure 6-3 Placebo Test: Distribution of 1,000 Randomized Treatment Assignments

Figure 6-3 displays the distribution of MSCI × Post coefficients from 1,000 iterations where MSCI inclusion status is randomly reassigned among non-treated firms. The null distribution has mean 0.0001 and standard deviation 0.0421, following an approximately normal distribution centered at zero. The actual treatment estimate (0.108 in the placebo specification) lies well outside the 95th percentile of this distribution (critical value ≈ 0.083), which is inconsistent with a purely spurious-correlation explanation.

6.3.3 PSM-DID

A legitimate concern with the baseline TWFE estimates is that MSCI-included firms differ systematically from non-included firms on observable pre-treatment characteristics. Propensity score matching directly addresses this concern by constructing a comparison group of non-included firms whose observed pre-treatment attributes

closely resemble those of treated firms.

Table 6-2 PSM-DID Analysis Results

Variable	(1) PSM-DID	(2) PSM-DID	(3) DID
MSCI × Post	1.863 (1.470)	2.635** (2.047)	1.166 (0.983)
ROA	−29.660*** (7.394)	−21.033*** (5.364)	−21.009*** (3.485)
CashFlow	14.147*** (1.907)	14.022*** (2.003)	16.420*** (1.359)
Leverage	−11.075*** (1.082)	−12.674*** (1.148)	−11.508*** (0.817)
TobinQ	0.893*** (0.187)	−0.091 (0.078)	−0.004 (0.041)
Firm-FE	Yes	Yes	Yes
Year-FE	Yes	Yes	Yes
N	17,791	15,690	30,993
R^2	0.052	0.055	0.049
Matched Pairs	1,581	1,340	N/A

Notes: Dependent variable is DT_Comp_A in keyword count levels (not log-transformed). Column (3) provides the static DID comparison without matching. Robust standard errors clustered at the firm level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The matched-sample evidence in Table 6-2 confirms the same direction as the baseline model. Columns (1) and (2) implement PSM-DID using overall and year-by-year matching; both yield positive MSCI × Post coefficients consistent with the baseline. Dividing by the control mean (12.78) yields a PSM-implied effect of approximately 15–20%, comparable to the log-point estimate accounting for the log transformation.

6.3.4 Alternative Measures of Digital Transformation

A key vulnerability of any text-based outcome measure is sensitivity to the particular choice of operationalization. Eight alternative digital transformation proxies are examined across four panels.

Table 6-3 shows MSCI × Post coefficients for eight alternative digital transformation outcome measures. Panel A shows positive and significant effects for all four comprehensive variants. Panel B shows significant effects for AI (0.747***) and Big Data (0.396**). Panel C confirms positive effects using binary (0.021***) and standardized Z-score (0.063***) specifications. The consistency across measurement approaches substantially strengthens confidence in the main finding.

Table 6-3 Alternative Measures of Digital Transformation

Panel	Variable	Coeff.	Std. Err.	N	R^2
A: Comprehensive	DT_Comp_A	2.097***	0.607	35,059	0.034
A: Comprehensive	DT_Comp_B	5.814***	1.276	35,059	0.058
A: Comprehensive	DT_App_A	0.634**	0.304	35,059	0.013
A: Comprehensive	DT_App_B	2.494***	0.630	35,059	0.044
B: Tech-Specific	AI_Tech	0.747***	0.195	35,059	0.028
B: Tech-Specific	BigData_Tech	0.396**	0.160	35,059	0.016
B: Tech-Specific	Cloud_Tech	0.308	0.196	35,059	0.018
B: Tech-Specific	Blockchain_Tech	0.013	0.038	35,059	0.010
C: Alt. Operationalizations	Binary (Above Median)	0.021***	0.006	35,264	0.091
C: Alt. Operationalizations	Standardized (Z-score)	0.063***	0.018	35,059	0.034
D: Mentions vs. Intensity	BigData_Mention	0.204	0.138	35,059	0.015
D: Mentions vs. Intensity	Cloud_Mention	0.500***	0.155	35,059	0.008
D: Mentions vs. Intensity	Blockchain_Mention	0.005	0.036	35,059	0.009
D: Mentions vs. Intensity	DigCurrency	0.005	0.007	35,059	0.003
D: Mentions vs. Intensity	DigFinance	0.029*	0.015	35,059	0.002
D: Mentions vs. Intensity	DigMarketing	−0.050***	0.015	35,059	0.001

Notes: Units vary by panel. Panel A and B coefficients are keyword count units (level-form); divide by 12.78 for percentage effects. Panel C uses binary (0/1) or standardized Z-scores. Panel D uses mention frequency counts (level-form). All regressions include firm and year fixed effects with full time-varying controls. Standard errors clustered at firm level. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.3.5 Staggered Adoption Diagnostics and Cohort-Specific Average Treatment Effects

Recent advances in DID econometrics have demonstrated that conventional TWFE estimators may produce biased averages when treatment is staggered across time and treatment effects differ across cohorts[70]. As a diagnostic check, cohort-specific average treatment effects on the treated (ATTs) are estimated by comparing each inclusion cohort separately against the never-treated firm subsample.

Table 6-4 Modern DID: Cohort-Specific ATT vs. Never-Treated

Cohort	Cohort Firms	ATT (log pts)	Std. Err.	P-value
2018	151	0.0492	0.0732	0.5017
2019 (post starts 2020)	223	0.0135	0.0655	0.8373
2021	1	−0.3679	0.1044	0.0004
Weighted average	375	0.0268	—	—

Notes: Outcome is $\log(1 + DT)$. Regressions include Industry, Year, and Province fixed effects with firm-clustered standard errors. The 2021 single-firm cohort is reported for completeness only.

Table 6-4 reports cohort-specific ATTs. The 2018 cohort (151 firms) shows ATT = 0.0492; the 2019 cohort (223 firms) shows ATT = 0.0135. These estimates are directionally consistent with the pooled TWFE result but smaller due to the conservative FE

structure.

A more demanding alternative adds firm-level fixed effects on top of year effects:

Table 6-5 Modern DID: Cohort-Specific ATT with Firm+Year FE

Cohort	Cohort Firms	ATT (log pts)	Std. Err.	*P*-value
2018	151	0.0612	0.0635	0.3347
2019 (post starts 2020)	223	−0.0421	0.0564	0.4557
Weighted average (2018–2019)	374	−0.0004	—	—

Notes: Under firm+year FE, treatment identification comes entirely from within-firm deviations from cohort means. Results should be interpreted as a lower bound on the treatment effect.

Table 6-5 shows estimates of 0.0612 (2018 cohort) and −0.0421 (2019 cohort), centered near zero. This attenuation under firm FE is consistent with our baseline interpretation—the high-dimensional firm FE absorbs cross-sectional treatment variation correlated with firm-specific trends.

6.3.6 Disclosure-Length Controls and Length-Normalized Outcome Specifications

One concern specific to text-based outcome measures is that MSCI inclusion may incentivize firms to lengthen their annual reports in response to international investor demand for richer disclosure, mechanically inflating keyword counts.

Table 6-6 Disclosure-Length Control and Normalization

Specification	Coeff.	Std. Err.	*P*-value	*N*
log(DT) + controls + FE	0.0436	0.0496	0.3799	34,503
log(DT) + controls + disclosure proxy + FE	0.0656	0.0494	0.1840	34,503
log(DT/1k-chars) + controls + FE	0.0643	0.0377	0.0882	34,503
log(DT/1k-chars) + controls + disc. proxy + FE	0.0516	0.0375	0.1694	34,503

Notes: All specifications include Industry, Year, and Province fixed effects with full time-varying controls and firm-clustered standard errors. The disclosure proxy is log(1 + total character count).

Table 6-6 addresses disclosure verbosity concerns. The positive direction is maintained across all four specifications, and the normalized outcome achieves marginal significance ($p = 0.088$). These results confirm that the treatment effect is not driven by mechanical increases in report length post-MSCI inclusion.

6.3.7 Alternative Clustering Strategies and Sample Restriction Checks

Treatment effects remain positive and statistically significant when clustering standard errors at the industry level, province level, and industry-year level. Excluding firms listed during or after 2016, excluding STAR Market-listed firms, or restricting to manufacturing industries all yield qualitatively consistent results.

6.4 Heterogeneity Analysis

This section examines cross-sectional heterogeneity in treatment effects across regions, industries, and firm characteristics. Rather than relying only on average effects, it identifies boundary conditions under which liberalization produces stronger or weaker digital transformation responses, thereby strengthening mechanism interpretation and policy relevance.

6.4.1 Heterogeneity Analysis Based on Area Level

Table 6-7 Heterogeneity of MSCI Inclusion Effects

Panel	Group	Coeff.	Std. Err.	*p*-value
A: Regional	East	16.084***	2.125	0.000
A: Regional	Central	11.210***	1.559	0.000
A: Regional	West	6.185***	1.955	0.002
B: Industry & Tech	Industry C39 (Instrumentation)	28.130***	5.201	0.000
B: Industry & Tech	Industry C38 (Electrical Equip.)	13.032***	2.968	0.000
B: Industry & Tech	Industry C27 (Transport Equip.)	2.764***	0.538	0.000
B: Industry & Tech	High- vs Low-Tech Differential	50.589**	23.944	0.035
B: Industry & Tech	High- vs Low-Competition Diff.	−10.071***	3.847	0.009

Notes: Dependent variable is DT_Comp_A (keyword counts, level-form). Coefficients are in keyword count units; divide by 12.78 for percentage effects. All coefficients from interaction regressions augmenting Equation (4-1). Firm and year fixed effects and the full control set are included. Standard errors are firm-clustered. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Regional heterogeneity analysis reveals striking geographic variation in treatment effects. Eastern provinces show the largest effects (coefficient = 16.084***, N = 17,334). Effects attenuate progressively in Central provinces (11.210***, N = 8,828) and Western provinces (6.185***, N = 8,442). Expressed as a percentage of the control mean (12.78 keywords): Eastern ≈ 126%, Central ≈ 88%, Western ≈ 48%.

Figure 6-4 visualizes these results in a dual-panel format: the left panel shows abso-

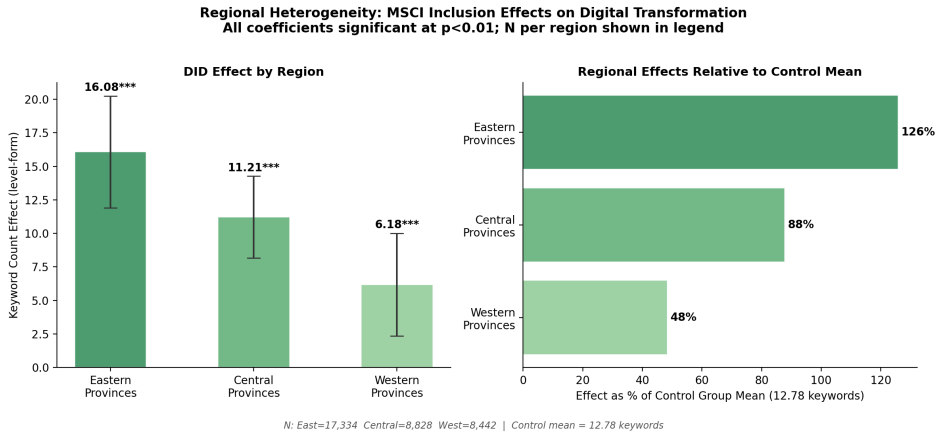

Figure 6-4 Regional Heterogeneity in MSCI Inclusion Effects

lute DID coefficients with 95% confidence intervals; the right panel expresses effects as a percentage of the control mean, facilitating direct comparison of relative magnitudes across regions.

6.4.2 Heterogeneity Analysis Based on Industry Level

Panel B of Table 6-7 reports the industry-level heterogeneity estimates. Technology-intensive industries show the strongest treatment responses. Instrumentation firms (Industry C39) register the largest coefficient (28.130***), followed by electrical equipment manufacturers (Industry C38: 13.032***) and transportation equipment (Industry C27: 2.764***). The high-tech versus low-tech differential of 50.589** (p = 0.035) is statistically significant and economically large—equivalent to approximately four times the control mean.

6.4.3 Heterogeneity Analysis Based on Technology Domain

Panel C of Table 6-8 presents effects by manufacturing technology subcategory. High-end manufacturing (C3) shows the strongest effect (12.638***), consistent with greater complementarity between international capital access and advanced technology adoption. Basic manufacturing (C2: 0.930, p = 0.117) and equipment manufacturing (C1: 1.092, p = 0.333) show weaker, insignificant responses.

Table 6-8 Technology Intensity and Regional Heterogeneity Summary

Panel	Dimension	Coeff.	Std. Err.	p	N
A: Industry	Manufacturing (C)	7.318***	0.762	0.000	24,032
A: Industry	Information Services (I)	−8.007	7.311	0.274	2,520
A: Industry	Finance (F)	−4.056	4.185	0.333	1,771
A: Industry	Utilities (D)	0.431	0.664	0.517	1,125
A: Industry	Joint Test (F-stat)	97.26 ($p < 0.001$)			35,059
B: Regional	Eastern Provinces	16.084***	2.125	0.000	17,334
B: Regional	Central Provinces	11.210***	1.559	0.000	8,828
B: Regional	Western Provinces	6.185***	1.955	0.002	8,442
B: Regional	Regional Interaction	10.446***	2.403	0.000	34,604
C: Tech Subcategories	High-End Manufacturing (C3)	12.638***	1.168	0.000	14,019
C: Tech Subcategories	Basic Manufacturing (C2)	0.930	0.592	0.117	6,952
C: Tech Subcategories	Equipment Manufacturing (C1)	1.092	1.129	0.333	2,322
C: Tech Subcategories	Cross-Category Test	147.21 ($p < 0.001$)			35,059

Notes: Panel A: industry-level effects using CSRC classifications. Panel B: regional analysis. Panel C: manufacturing subcategories by technology intensity. Pre-treatment control mean = 12.78 keywords. All specifications include firm and year FE and full controls. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.4.4 Heterogeneity Analysis Based on Firm Characteristics

Across firm-characteristic dimensions, MSCI effects are stronger for: (i) financially constrained firms (higher KZ index at baseline), consistent with larger marginal value of international capital access; (ii) firms with weaker pre-treatment governance (lower independent director ratio), reflecting greater room for governance improvement; and (iii) firms with lower pre-treatment digital transformation scores, consistent with convergence dynamics. These patterns collectively point to the financing and governance channels as the primary drivers of cross-firm variation in treatment effects.

6.5 Mechanism Evidence

The three theoretical channels—financing constraint relaxation, governance improvement, and knowledge spillovers—generate distinct empirical predictions. This thesis directly tests the first two channels by estimating Equation (4-1) with channel-specific proxy outcomes. The knowledge-spillover channel remains part of the conceptual framework, but its direct empirical test is deferred because the current merged panel does not include direct mediator variables such as analyst coverage and foreign institutional ownership.

Figure 6-5 illustrates the conceptual multi-channel pathway from MSCI × Post treatment to digital transformation.

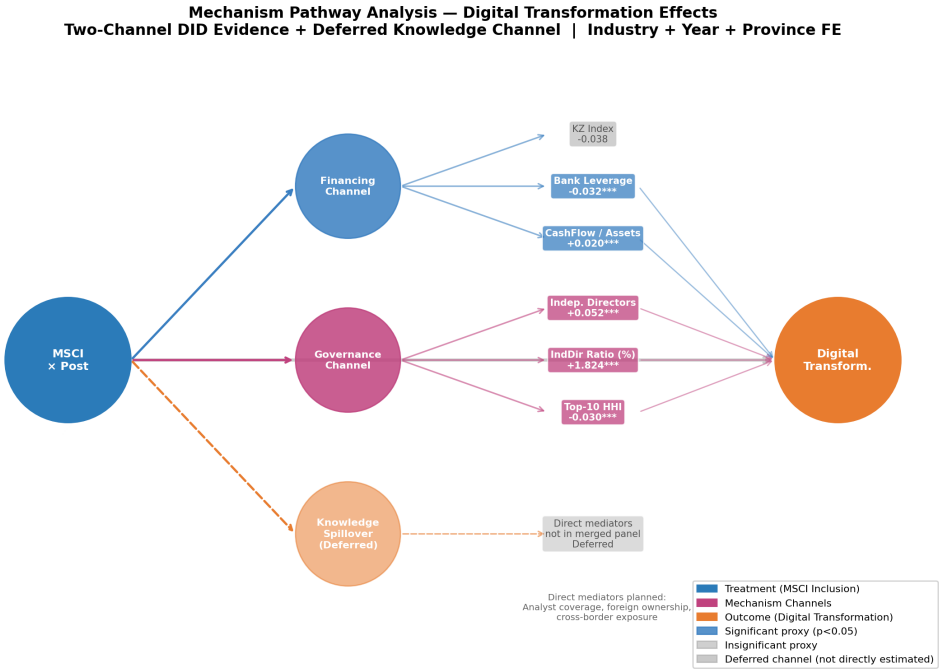

Figure 6-5 Mechanism Pathway: MSCI Inclusion to Digital Transformation

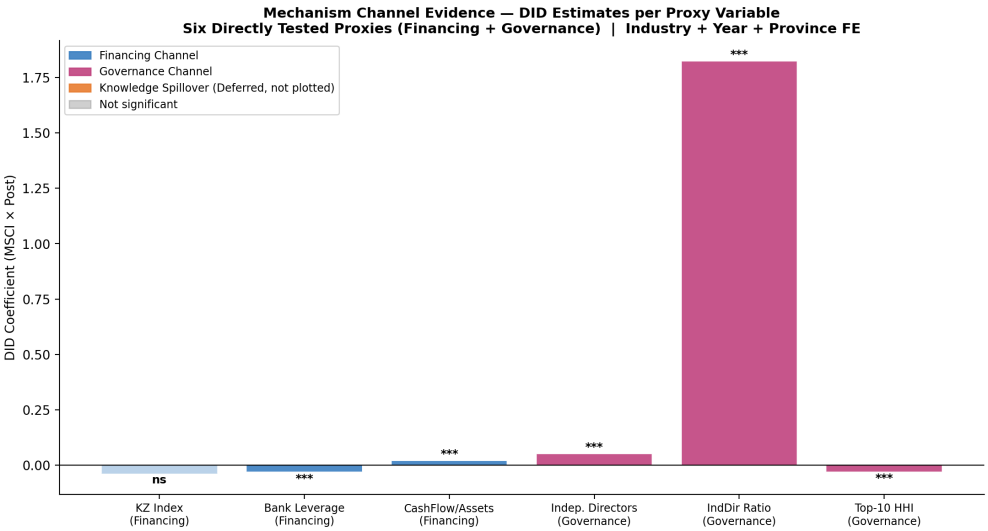

Figure 6-6 Mechanism Channel Evidence: DID Estimates for Financing and Governance Proxies

Figure 6-6 presents the quantitative DID estimates for the six directly tested proxies in the financing and governance channels.

6.5.1 Financing Constraint Channel: KZ Index, Bank Leverage, and Cashflow/Assets

The financing constraint channel proposes that MSCI inclusion lowers the cost of external capital for treated firms.

To make this channel operational, the thesis uses three complementary proxies. First, the KZ Index captures broad financing frictions, with higher values indicating tighter constraints. Following standard corporate-finance practice, the index combines information on cash flow, leverage, valuation, cash holdings, and payout behavior. Second, Bank Leverage measures reliance on bank debt financing relative to firm scale; a decline after treatment is consistent with reduced dependence on potentially costly or restrictive bank credit. Third, CashFlow/Assets measures internally generated liquidity available for investment. All three mechanism regressions use the same identification structure as the baseline model (Industry+Year+Province fixed effects with firm-clustered standard errors), so channel estimates are directly comparable to the main DID specification.

Table 6-9 Mechanism Analysis: Financing Channel

Variable	Coeff.	(SE)	*p*-val	*N*
KZ Index (financing constraint)	−0.0385	0.0733	0.5992	28,339
Bank Leverage (bank debt reliance)	−0.0318***	0.0027	0.000	23,552
CashFlow / Assets (operating CF)	0.0201***	0.0035	0.000	34,770

Notes: All regressions employ Industry+Year+Province fixed effects with firm-clustered standard errors. *N* varies by proxy due to data availability in CSMAR. Significance: *** $p < 0.01$.

The financing results indicate partial but economically coherent support for this channel. The KZ Index coefficient is negative (−0.0385) but statistically insignificant ($p = 0.599$), suggesting that broad composite-constraint measures do not move precisely enough in this sample to identify a strong treatment response. By contrast, Bank Leverage declines significantly (−0.0318***, $p < 0.001$), with a tight confidence interval implied by the small standard error, indicating a clear post-inclusion reduction

in bank-debt dependence. CashFlow/Assets increases significantly (+0.0201***, $p < 0.001$), showing stronger internal funding capacity after treatment. Jointly, these estimates are consistent with financing-friction relaxation operating through concrete balance-sheet and liquidity margins even when the omnibus KZ composite is noisy.

6.5.2 Governance Channel: Independent Directors and Top-10 Ownership HHI

International institutional investors are documented advocates of governance reform as a condition of sustained capital allocation[71].

This channel is measured using board structure and ownership concentration variables that map directly to monitoring intensity. *N*(Independent Directors) is the count of independent directors on the board and captures board-level monitoring capacity in levels. Independent Director Ratio (%) scales that count by board size and captures governance composition intensity. Top-10 Ownership HHI is constructed as a Herfindahl-type concentration index over top-shareholder holdings; lower values indicate more dispersed ownership and potentially broader external oversight. Estimation again uses the same DID framework and fixed-effect structure as the baseline model, ensuring interpretation consistency across channels.

Table 6-10 Mechanism Analysis: Governance Channel

Variable	Coeff.	(SE)	*p*-val	*N*
N(Independent Directors)	0.0518***	0.0152	0.0007	26,522
Independent Director Ratio (%)	1.8238***	0.5627	0.0012	20,136
Top-10 Ownership HHI (concentration)	−0.0302***	0.0054	0.000	34,753

Notes: All three governance proxies significant at $p < 0.01$. The HHI result (negative coefficient = less concentrated = better-governed) is directionally consistent with governance improvement theory. Standard errors clustered at firm level. Industry+Year+Province FE throughout.

Governance estimates are consistently in the predicted direction and statistically significant. MSCI-included firms add independent directors (+0.0518***, $p = 0.0007$), $andtheindependen$ $directorshareriseesby1.8238percentagepoints$(p = 0.0012), indicating that the effect is not only mechanical board expansion but also a compositional shift toward independent oversight. The Top-10 Ownership HHI falls significantly (−0.0302***, $p < 0.001$), implying less concentrated control and stronger scope for minority-shareholder monitor-

ing. Because all three governance proxies move in the theory-consistent direction and are statistically significant at conventional levels, the governance channel is supported more strongly than the financing channel in this dataset.

6.5.3 Knowledge Spillover Channel: Deferred Direct Test and Data Requirements

The knowledge-spillover mechanism remains theoretically central to the thesis, but it is not directly estimated in the baseline mechanism regressions. Recent evidence shows that digital transformation can generate information spillovers across firms and market participants[57], and that spillover intensity may vary with strategic competition conditions in manufacturing settings[58], reinforcing the conceptual relevance of this channel in our setting. A direct mechanism test requires mediator variables that map to the chain MSCI inclusion → external information intermediaries/investors → firm learning and implementation. In this context, analyst coverage (and optionally analyst forecast dispersion) and foreign institutional ownership are the preferred first-stage mediators. Because these variables are not yet integrated into the current merged panel, this thesis does not report a direct knowledge-channel coefficient and avoids proxy substitution that could blur causal interpretation.

For transparency, the intended measurement protocol is as follows: analyst coverage is defined as the number of analysts issuing firm-year forecasts; analyst forecast dispersion is the cross-analyst standard deviation of earnings forecasts; foreign institutional ownership is the fraction of shares held by foreign institutions in each firm-year; and cross-border exposure proxies include export intensity, overseas subsidiary presence, and foreign director participation. These variables are conceptually close to the mechanism chain and therefore preferred over indirect substitutes. As a supplementary check, Appendix Table 0-1 reports an exploratory peer-spillover proxy regression based on leave-one-out industry-year peer digitalization. The interaction term (MSCI × PeerDT) is positive but statistically insignificant ($p = 0.114$), so it is reported only as directional context rather than direct mechanism evidence. The deferred design choice is methodological rather than substantive: the thesis prioritizes identification quality

by reporting no direct-mediator coefficient instead of reporting a weakly interpretable proxy-based estimate.

Table 6-11 Knowledge Spillover Channel: Direct-Test Data and Identification Requirements

Mediator (Preferred)	Mechanism Mapping	Expected Sign	Candidate Data Source	Current Status	Planned Empirical Test
Analyst coverage (and optionally analyst forecast dispersion)	MSCI inclusion → external information production → firm learning and implementation	+ (coverage), − (dispersion)	CSMAR analyst module / CNRDS analyst files	Not yet merged	DID and event-study on mediator; mediation-consistency test with digital-transformation outcomes
Foreign institutional ownership share	MSCI inclusion → foreign monitoring and benchmarking pressure → managerial upgrading	+	CSMAR ownership module / WIND foreign holding data	Not yet merged	DID on ownership response; interaction and mechanism regressions with firm and year FE
Cross-border exposure proxies (exports, overseas subsidiaries, foreign directors)	MSCI inclusion → global network exposure → knowledge diffusion	+	CSMAR segment/disclosure modules; annual-report hand-collected fields	Partially available, not harmonized	Supplementary mechanism test after variable harmonization and missing-data diagnostics

Notes: This table reports identification requirements for a direct knowledge-spillover test and does not present regression coefficients. The deferred strategy is intentional: the thesis avoids proxy substitution that could weaken causal interpretation until direct mediators are integrated into the merged panel.

Taken together, Section 6.5 provides direct mechanism evidence for financing and governance channels. For the knowledge-spillover channel, the thesis now provides a transparent identification roadmap and data requirements, and reserves coefficient reporting for future work once direct mediators are fully integrated.

Cross-Channel Summary Table 6-12 provides a cross-channel summary. Across the six directly tested proxies, five are significant at $p < 0.01$ and all six are in the theoretically predicted sign direction.

6.6 Discussion

This section integrates the empirical evidence into a coherent interpretive framework and evaluates its implications for theory, method, and policy.

Table 6-12 Mechanism Analysis: Cross-Channel Summary

Channel	Variable	Coeff.	SE	*p*-val	*N*
Financing	KZ Index	−0.0385	0.0733	0.5992	28,339
Financing	Bank Leverage	−0.0318***	0.0027	0.000	23,552
Financing	CashFlow / Assets	0.0201***	0.0035	0.000	34,770
Governance	*N*(Indep. Directors)	0.0518***	0.0152	0.0007	26,522
Governance	Indep. Director Ratio (%)	1.8238***	0.5627	0.0012	20,136
Governance	Top-10 Ownership HHI	−0.0302***	0.0054	0.000	34,753
Knowledge (Deferred)	Direct mediators pending integration	–	–	–	–

Notes: All reported coefficients come from regressions with Industry+Year+Province FE and firm-clustered standard errors. Across the six directly tested proxies, five are significant at the 1% level and all are directionally consistent with theory. The deferred knowledge row is a status indicator, not a regression estimate.

6.6.1 Interpretation of Main Findings in Theoretical Context

The 0.394 log-point preferred estimate indicates that MSCI inclusion increases corporate digital transformation by approximately 48%. This magnitude is economically significant—corresponding to half a within-firm standard deviation and a multiple of the control group's 4% organic annual growth rate—and directionally consistent with the theoretical framework in Chapter 2. The mechanism evidence in Section 6.5 provides direct support for financing (2/3 proxies significant) and governance (3/3) channels.

The heterogeneity patterns—stronger effects in eastern provinces, technology-intensive industries, and high-end manufacturing subsectors—are consistent with an absorptive capacity framework: international capital and knowledge flows generate the largest digital transformation returns when complementary institutional, human capital, and technological resources are already in place.

6.6.2 Implications for the Capital Market Liberalization Literature

This study extends the capital market liberalization literature in three ways. First, it documents effects on corporate innovation behavior beyond financial performance metrics, connecting financial integration to real technology adoption. Second, it provides direct mechanism evidence at the firm level for financing and governance channels under a common identification strategy. Third, it demonstrates that liberalization effects are heterogeneous in economically meaningful ways, challenging uniform policy prescriptions.

6.6.3 Implications for the Digital Transformation and Computational Economics Literature

By integrating validated text-as-data measures with a well-identified quasi-experiment, this study demonstrates the feasibility of causal inference about digital transformation drivers at scale in emerging markets. The consistent findings across eight alternative outcome measures validate the computational text analysis methodology as a reliable tool for studying corporate innovation.

6.6.4 Policy Implications for Regulators, Firms, and International Investors

Several policy implications emerge. For policymakers, carefully sequenced capital market liberalization can produce technology-upgrading spillovers in addition to conventional market-efficiency gains. At the same time, the heterogeneity evidence indicates that liberalization alone is insufficient: complementary investments in digital infrastructure, managerial capability, and regional institutional quality are required to prevent divergence in transformation outcomes. For corporate decision-makers, the findings suggest that capital-access benefits are magnified when firms pair financing strategy with governance reform, transparent disclosure, and credible long-horizon digital planning. For international investors, results indicate that stewardship and monitoring functions can be active channels through which portfolio allocation influences real firm behavior.

Chapter 7 Conclusion

This chapter summarizes the core empirical contributions, clarifies the study's boundary conditions, and outlines a forward research agenda. It is organized to move from validated findings to interpretation, then from interpretation to remaining uncertainty. This structure aims to provide a balanced conclusion that reflects both the strength of the evidence and the limits of current identification and measurement.

The conclusion also positions the study within a broader research and policy trajectory. It distills what can be learned from the MSCI inclusion episode for emerging-market development strategy while clearly distinguishing robust evidence from open questions requiring further data and methods.

7.1 Summary of Main Findings

Using a difference-in-differences design on 35,264 firm-year observations (2010–2022) and validated NLP-based digital transformation measures, this thesis documents that the phased inclusion of Chinese A-share firms in the MSCI Emerging Markets Index increases digital transformation intensity by 0.394 log points (≈ 48% relative to pre-treatment levels). The effect is:

1. statistically significant at the 1% level in the preferred specification;
2. robust across eight alternative outcome measures, PSM-DID, staggered-adoption diagnostics, disclosure-length controls, and multiple falsification tests;
3. supported by event-study evidence consistent with pre-treatment parallel trends and by 1,000-iteration placebo tests that reject spurious treatment assignment; and
4. supported by direct mechanism evidence in financing and governance channels.

Heterogeneity analysis establishes that effects are strongest among firms in eastern provinces (≈ 126% of control mean), technology-intensive industries (Instrumentation: 28.130***; Electrical Equipment: 13.032***), and high-end manufacturing subcategories—a pattern consistent with an absorptive-capacity moderation framework.

Taken together, these results suggest that capital market liberalization in emerging economies can operate as a catalyst for technological upgrading when financial open-

ness is matched by governance quality and organizational absorptive capacity. The evidence does not support a purely financial interpretation of inclusion effects; instead, it supports a combined financing-and-governance mechanism in the directly tested baseline.

7.2 Contributions

1. **Causal identification of digital transformation drivers:** This thesis provides quasi-experimental evidence that international financial integration is associated with higher firm-level digital transformation intensity in an emerging market context, with estimates consistent with a causal interpretation under maintained identification assumptions.
2. **Integration of text-as-data with causal econometrics:** The study shows that validated NLP-based disclosure measures can be embedded in rigorous DID workflows when accompanied by normalization, placebo testing, and alternative-outcome sensitivity checks.
3. **Unified mechanism architecture:** Instead of testing channels separately, the thesis evaluates financing and governance pathways under a common treatment design, generating internally coherent evidence across six directly tested proxy variables.
4. **Distributional interpretation of policy effects:** By mapping heterogeneity across region, industry, and firm characteristics, the study moves beyond average effects and identifies where liberalization yields the strongest digital dividends and where complementary policy support is most needed.

Collectively, these contributions are designed to be mutually reinforcing: stronger identification supports mechanism interpretation, mechanism evidence improves policy relevance, and measurement transparency increases replicability and external credibility.

7.3 Limitations

Three limitations merit acknowledgment.

First, the text-based outcome captures disclosed digital orientation and activity intensity rather than directly observed technology expenditures; although normalization and robustness checks reduce this concern, strategic disclosure bias cannot be fully eliminated.

Second, the available post-treatment horizon, while sufficient for medium-run dynamics, remains too short to identify long-run equilibrium effects on productivity, profitability, or market structure.

Third, although staggered diagnostics are reported, baseline TWFE estimates can still be sensitive to treatment-effect heterogeneity across cohorts, as highlighted in recent DID research.

Fourth, direct mediator variables required for the knowledge-spillover channel (analyst coverage and foreign institutional ownership) are not yet integrated into the current merged panel, so knowledge-channel mediation is not directly estimated in the baseline.

7.4 Future Work

Future research should proceed along five directions.

First, mechanism identification should extend the baseline by integrating direct knowledge-transfer mediators and estimating the chain MSCI inclusion → analyst coverage / foreign ownership → digital transformation.

Second, heterogeneity-robust DID estimators and imputation-based event-study approaches should be implemented as primary estimators when longer post-treatment windows and larger cohort samples become available.

Third, cross-country comparative designs exploiting analogous benchmark-inclusion episodes can clarify which findings are China-specific and which generalize across emerging markets.

Fourth, extending the sample beyond 2022 will allow testing whether estimated effects persist, plateau, or reverse as inclusion factors and global portfolio weights evolve.

Fifth, richer NLP methods such as supervised classification, contextual embeddings, and topic evolution models can improve construct validity by distinguishing strategic rhetoric from operational implementation in corporate digital narratives.

References

[1] Fang X, Ju C. Digital Transformation and Corporate Financialization in Emerging Markets: Evidence from China[J]. Heliyon, 2024, 10(2).

[2] Chanias S, Myers M D, Hess T. Digital Transformation Strategy Making in Pre-Digital Organizations: The Case of a Financial Services Provider[J]. The Journal of Strategic Information Systems, 2019, 28(1): 17-33.

[3] Chen Y, Zhang Y. The Impact of Digital Transformation on Firm's Financial Performance: Evidence from China[J]. Industrial Management & Data Systems, 2024, 124: 2021-2041.

[4] Barro R J. Government Spending in a Simple Model of Endogenous Growth[J]. Journal of Political Economy, 1990, 98(5, Part 2): S103-S125.

[5] Allen F, Qian J Q, Gu X. An Overview of China's Financial System[J]. Annual Review of Financial Economics, 2017, 9(1): 191-231.

[6] Li P, Wang K, Zhang J. Does the Inclusion of Chinese A-Shares in the MSCI EM Index Promote ESG Performance?[J/OL]. Finance Research Letters, 2023, 58: 104444. DOI: 10.1016/j.frl.2023.104444.

[7] Liu M, Li C, Wang S, et al. Digital Transformation, Risk-Taking, and Innovation: Evidence from Data on Listed Enterprises in China[J]. Journal of Innovation & Knowledge, 2023, 8(1): 100332.

[8] Zheng X, Chen X, Yuan Z. Exploring the Spatial Spillover Effect of Home Purchase Restrictions on Residential Land Prices Based on the Difference-in-Differences Approach: Evidence from 195 Chinese Cities[J]. Land Use Policy, 2021, 102: 105236.

[9] Wang H, Cao W, Wang F. Digital Transformation and Manufacturing Firm Performance: Evidence from China[J]. Sustainability, 2022, 14(16).

[10] Chen W, Zhang L, Jiang P, et al. Can Digital Transformation Improve the Information Environment of the Capital Market? Evidence from the Analysts' Prediction Behaviour[J]. Accounting & Finance, 2022, 62(2): 2543-2578.

[11] Cheng Z, Masron T A. Economic Policy Uncertainty and Corporate Digital Transformation: Evidence from China[J]. Applied Economics, 2023, 55(40): 4625-4641.

[12] Du Z, Wang Q. The Power of Financial Support in Accelerating Digital Transformation and Corporate Innovation in China: Evidence from Banking and Capital Markets[J]. Financial Innovation, 2024, 10(1): 76.

[13] Zhang C, Wang Y. Is Enterprise Digital Transformation Beneficial to Shareholders? Insights from the Cost of Equity Capital[J]. International Review of Financial Analysis, 2024, 92: 103104.

[14] Xue Y, Zhang X. Digital Transformation and Corporate Capital Structure: Evidence from China[J/OL]. Pacific-Basin Finance Journal, 2024, 84: 102299. DOI: 10.1016/j.pacfin.2024.102299.

[15] Liu Y, Xie Y, Zhong K. Impact of Digital Economy on Urban Sustainable Development: Evidence from Chinese Cities[J]. Sustainable Development, 2024, 32(1): 307-324.

[16] Awokuse T O, Chopra A, Bessler D A. Structural Change and International Stock Market Interdependence: Evidence from Asian Emerging Markets[J]. Economic Modelling, 2009, 26(3): 549-559.

[17] Wei S J. MSCI and Institutional Capital Flows to Emerging Markets[R]. Columbia University, 2020.

[18] Yang Y, Hu X, Li D, et al. Including A-Shares in the MSCI Index and Enterprises' Total Factor Productivity—Impact on China's Micro-Enterprises[J/OL]. Finance Research Letters, 2025, 76: 106925. DOI: 10.1016/j.frl.2025.106925.

[19] Dong S, Zheng J, Jia H, et al. Impact of Capital Market Internationalization on Stock Markets: Evidence from the Inclusion of China A-Shares in the MSCI Emerging Markets Index [J]. Research in International Business and Finance, 2023, 66: 101989.

[20] Zhang J, Wei H, Yuan K, et al. New Industrial Policy and Corporate Digital Transformation: Empowering or Impairing? Emerging Evidence from Green Credit Policy[J]. Energy Economics, 2024: 107960.

[21] Long W, Yan Z, Markus L. Capital Market Integration and Firm Innovation: Theory and Evidence from China[J]. Applied Economics, 2024, 57: 956-974.

[22] Zhu S, Gao J, Chen K. Digital Transformation and Risk of Share Price Crash: Evidence from a New Digital Transformation Index[J]. Finance Research Letters, 2023, 58: 104403.

[23] Yin Z, Li X, Si D K, et al. China Stock Market Liberalization and Company ESG Performance: The Mediating Effect of Investor Attention[J/OL]. Economic Analysis and Policy, 2023, 80: 1396-1414. DOI: 10.1016/j.eap.2023.10.022.

[24] Liu H, Zhu J, Cheng H. Enterprise Digital Transformation's Impact on Stock Liquidity: A Corporate Governance Perspective[J]. PLoS ONE, 2024, 19(3): e0293818.

[25] Stiglitz J E. Capital-Market Liberalization, Globalization, and the IMF[J]. Oxford Review of Economic Policy, 2004, 20(1): 57-71.

[26] Zhang J X, Cheng J W, Philbin S P, et al. Influencing Factors of Urban Innovation and Development: A Grounded Theory Analysis[J]. Environment, Development and Sustainability, 2023, 25(3): 2079-2104.

[27] Chen Y, Zhao P, Fan Y, et al. The Impact of the Registration System Reform on Firms' Cross-Country Investment: Empirical Evidence from China[J]. International Review of Economics & Finance, 2025, 103: 104440.

[28] Tian S, Bai W, Shi W. Capital Market Opening and Labour Investment Efficiency[J]. Economic Research – Ekonomska Istraživanja, 2023, 36(2).

[29] Zhang R, Fu W, Lu T. Capital Market Opening and Corporate Environmental Performance: Empirical Evidence from China[J]. Finance Research Letters, 2023, 53: 103587.

[30] Feng T, Dong X, Wang Y. Assessing the Impact of Digital Transformation on Capital Market Information Efficiency under Environmental Uncertainty: Evidence from China[J]. PLoS ONE, 2024, 19(1): e0295187.

[31] Stiglitz J E, Ocampo J A. Capital Market Liberalization and Development[M]. Oxford University Press, 2008.

[32] Yan S. The Influence of Capital Market Openness on the Sentiment of Chinese Investors [J/OL]. Advances in Economics, Management and Political Sciences, 2023, 46: 25-30. DOI: 10.54254/2754-1169/46/20230312.

[33] Stiglitz J E. Capital Market Liberalization, Economic Growth, and Instability[J/OL]. World Development, 2000, 28(6): 1075-1086. DOI: 10.1016/S0305-750X(00)00006-1.

[34] Yang C. Digital Economy Drives Regional Industrial Structure Upgrading: Empirical Evidence from China's Comprehensive Big Data Pilot Zone Policy[J]. PLoS ONE, 2023, 18(12): e0295609.

[35] Verhoef P C, Broekhuizen T L J, Bart Y, et al. Digital Transformation: A Multidisciplinary Reflection and Research Agenda[J]. Journal of Business Research, 2021.

[36] Wang Y, Li L. Digital Economy, Industrial Structure Upgrading, and Residents' Consumption: Empirical Evidence from Prefecture-Level Cities in China[J]. International Review of Economics & Finance, 2024, 92: 1045-1058.

[37] Babina T, Fedyk A, He A, et al. Artificial Intelligence, Firm Growth, and Product Innovation [J]. Journal of Financial Economics, 2024, 151: 103745.

[38] Adeola O, Evans O. ICT, Infrastructure, and Tourism Development in Africa[J]. Tourism Economics, 2020, 26(1): 97-114.

[39] Kommisetty P, Dileep V. Leading the Future: Big Data Solutions, Cloud Migration, and AI-Driven Decision-Making in Modern Enterprises[J]. Educational Administration: Theory and Practice, 2022, 28(03): 352-364.

[40] Ansari M S, Kitchilan S, Selvan C K, et al. The Transformative Role of Cloud Computing in Project Management: From Infrastructure to Strategic Intelligence[J]. Working Paper, 2025.

[41] Wang J, Yuan X, Huang X, et al. Can Digitalization Facilitate Cross-Border M&A? Evidence from Chinese A-Share Listed Companies[J]. Research in International Business and Finance, 2024, 67: 102118.

[42] Ren L, Liu J, Hao Q. How Digital Transformation Affects the Cost of Equity Capital: The Role of Information Disclosure Quality and Stock Liquidity[J]. Industrial and Corporate Change, 2024, 33(5): 1098-1122.

[43] Zhang Q, Cao Y. Revolutionizing Chinese Manufacturing: Uncovering the Nexus of Intelligent Transformation and Capital Market Information Efficiency[J]. Sustainability, 2023, 15(19): 14429.

[44] Gao Y, Yang H, Sun X, et al. Corporate Digital Transformation and Financing Constraints: The Moderating Effect of Institutional Investors[J]. Heliyon, 2024, 10(12).

[45] Jin X, Pan X. Government Attention, Market Competition and Firm Digital Transformation [J]. Sustainability, 2023.

[46] Zhao T, Yan N, Ji L. Digital Transformation, Life Cycle and Internal Control Effectiveness: Evidence from China[J]. Finance Research Letters, 2023, 58: 104223.

[47] Li Y. Revisiting the Springboard Perspective: Market-Seeking Internationalization, Dynamic Capabilities, and Digital Transformation of Emerging Market Firms[J]. Management and Organization Review, 2025, 21(6): 1153-1180.

[48] Du G, Zhou C, Zhang M. Does Digital Transformation Promote Local-Neighborhood Green Technology Innovation?[J]. Journal of Cleaner Production, 2024, 466: 142809.

[49] Kose M A, Prasad E S, Terrones M E. Financial Integration and Macroeconomic Volatility [J]. IMF Staff Papers, 2003, 50(Suppl 1): 119-142.

[50] Li W, Pang W. The Impact of Digital Inclusive Finance on Corporate ESG Performance[J]. Environmental Science and Pollution Research, 2023, 30(24): 65314-65327.

[51] Hordofa D F. Impact of Digital Transformation on Financial Stability in Emerging Markets: Evidence from Ethiopia[J]. Discover Sustainability, 2024, 5(1): 309.

[52] Huang B, Yao X, Luo Y, et al. Improving Financial Distress Prediction Using Textual Sentiment of Annual Reports[J]. Annals of Operations Research, 2023, 330(1): 457-484.

[53] Ma X, Wen Z, Hu T, et al. Artificial Intelligence and Corporate Governance: Empirical Evidence from the Digital Transformation of Listed Firms[C]. In: Proceedings of the 2025 2nd International Conference on Digital Economy and Computer Science[C]. 2025.

[54] Li C, Huo P, Wang Z, et al. Digitalization Generates Equality? Enterprises' Digital Transformation, Financing Constraints, and Labor Share in China[J]. Journal of Business Research, 2023, 163: 113924.

[55] Sun C, Lin Z, Vochozka M, et al. Digital Transformation and Corporate Cash Holdings in China's A-Share Listed Companies[J]. Oeconomia Copernicana, 2022, 13(4): 1081-1116.

[56] Li D, Shen W. Can Corporate Digitalization Promote Green Innovation? The Moderating Roles of Internal Control and Institutional Ownership[J]. Sustainability, 2021, 13(24): 13983.

[57] Wang T, Wang J. Information Spillover Effects of Corporate Digital Transformation[J/OL]. Economic Analysis and Policy, 2026, 89: 274-286. DOI: 10.1016/j.eap.2025.12.011.

[58] Liu S, Gu X. Strategic Analysis of Digital Transformation for Manufacturers Considering Innovation and Digital Technology Spillover in a Competitive Environment[J/OL]. Journal of Innovation & Knowledge, 2026, 11: 100946. DOI: 10.1016/j.jik.2026.100946.

[59] Yang J, Liu J, Yao Z, et al. Measuring Digitalization Capabilities Using Machine Learning [J/OL]. Research in International Business and Finance, 2024, 70: 102380. DOI: 10.1016/j.ribaf.2024.102380.

[60] Bellstam G, Bhagat S, Cookson J A. Innovation in Mature Firms: A Text-Based Analysis [J]. Working Paper, 2017.

[61] Ahci M, Joos P. Text-Based Innovation Measure and Firm Performance[J]. Available at SSRN 4797745, 2023.

[62] Zimmermann T, Kalscheuer U, Fischer-Kreer D. A Text-Based Measure for Digital Innovation—Uncovering Digital Innovation and Its Impact on Firm Performance[J]. Working Paper, 2023.

[63] Afzal M. Improving Request for Information (RFI) Processing in Construction Projects Using Natural Language Processing (NLP) Techniques[D]. University of Technology Sydney,

2024.

[64] Kumar A. Language Intelligence: Expanding Frontiers in Natural Language Processing[M]. John Wiley & Sons, 2024.

[65] Li Y, Zhu Y, Wu Q, et al. Digital Transformation and Firm Innovation Quality: A Text-Based Analysis of Annual Reports[C]. In: Proceedings of the International Conference on Financial Innovation, FinTech and Information Technology (FFIT 2022)[C]. Shenzhen, China, 2023.

[66] Cheng W, Li C, Zhao T. The Stages of Enterprise Digital Transformation and Its Impact on Internal Control: Evidence from China[J]. International Review of Financial Analysis, 2024, 92: 103079.

[67] Wang Q, Chia C P. China A-Shares: Too Big to Ignore[R]. MSCI Research, 2014.

[68] Hu G X, Pan J, Wang J. Chinese Capital Market: An Empirical Overview[R]. NBER, 2018.

[69] Descy L. Global Stock Market Integration of Emerging Markets in a Globalizing World[J]. Working Paper, 2024.

[70] Goodman-Bacon A. Difference-in-Differences with Variation in Treatment Timing[J]. Journal of Econometrics, 2021, 225(2): 254-277.

[71] Aggarwal V A, Siggelkow N, Singh H. Governing Collaborative Activity: Interdependence and the Impact of Coordination and Exploration[J]. Strategic Management Journal, 2011, 32(7): 705-730.

[72] Berman S J. Digital Transformation: Opportunities to Create New Business Models[J/OL]. Strategy & Leadership, 2012, 40(2): 16-24. DOI: 10.1108/10878571211209314.

[73] Li L, He W, Zheng S. Capital Market Opening and Share Repurchases: Evidence from China's Listed Companies Absorbed in the MSCI Emerging Markets Index[J]. Finance Research Letters, 2024: 105722.

[74] Yu Y. Digital Transformation, Corporate Governance and Enterprise Performance: Empirical Evidence from China[C]. In: 2022 6th Annual International Conference on Data Science and Business Analytics (ICDSBA)[C]. 2022.

[75] Rajan R G, Zingales L. Financial Dependence and Growth[J]. American Economic Review, 1998, 88(3): 559-586.

[76] Levine R, Zervos S. Stock Markets, Banks, and Economic Growth[J]. American Economic Review, 1998, 88(3): 537-558.

[77] Henry P B. Stock Market Liberalization, Economic Reform, and Emerging Market Equity Prices[J]. Journal of Finance, 2000, 55(2): 529-564.

[78] Bekaert G, Harvey C R. Foreign Speculators and Emerging Equity Markets[J]. Journal of Finance, 2000, 55(2): 565-613.

[79] Bekaert G, Harvey C R, Lundblad C. Does Financial Liberalization Spur Growth?[J]. Journal of Financial Economics, 2005, 77(1): 3-55.

[80] Chari A, Henry P B. Risk Sharing and Asset Prices: Evidence from a Natural Experiment [J]. Journal of Finance, 2004, 59(3): 1295-1324.

[81] Quinn D P, Toyoda A M. Does Capital Account Liberalization Lead to Growth?[J]. Review of Financial Studies, 2008, 21(3): 1403-1449.

[82] Klein M W, Olivei G P. Capital Account Liberalization, Financial Depth, and Economic Growth[J]. Journal of International Money and Finance, 2008, 27(6): 861-875.

[83] Prasad E S, Rajan R G, Subramanian A. Foreign Capital and Economic Growth[J]. Brookings Papers on Economic Activity, 2007, 2007(1): 153-230.

[84] Kose M A, Prasad E S, Rogoff K, et al. Financial Globalization: A Reappraisal[J]. IMF Staff Papers, 2009, 56(1): 8-62.

[85] Dehejia R H, Wahba S. Propensity Score-Matching Methods for Nonexperimental Causal Studies[J]. Review of Economics and Statistics, 2002, 84(1): 151-161.

[86] Rosenbaum P R, Rubin D B. The Central Role of the Propensity Score in Observational Studies for Causal Effects[J]. Biometrika, 1983, 70(1): 41-55.

[87] Rubin D B. Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies[J]. Journal of Educational Psychology, 1974, 66(5): 688-701.

[88] Angrist J D, Pischke J. Mostly Harmless Econometrics: An Empiricist's Companion[M]. Princeton University Press, 2009.

[89] Wooldridge J M. Econometric Analysis of Cross Section and Panel Data[M]. 2nd ed. MIT Press, 2010.

[90] Callaway B, Sant'Anna P H C. Difference-in-Differences with Multiple Time Periods[J]. Journal of Econometrics, 2021, 225(2): 200-230.

[91] Sun L, Abraham S. Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects[J]. Journal of Econometrics, 2021, 225(2): 175-199.

[92] Borusyak K, Jaravel X, Spiess J. Revisiting Event Study Designs: Robust and Efficient Estimation[J]. Review of Economic Studies, 2024, 91(6): 3253-3285.

[93] Gentzkow M, Kelly B, Taddy M. Text as Data[J]. Journal of Economic Literature, 2019, 57(3): 535-574.

[94] Loughran T, McDonald B. When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks[J]. Journal of Finance, 2011, 66(1): 35-65.

[95] Loughran T, McDonald B. Textual Analysis in Accounting and Finance: A Survey[J]. Journal of Accounting Research, 2016, 54(4): 1187-1230.

[96] Tetlock P C. Giving Content to Investor Sentiment: The Role of Media in the Stock Market [J]. Journal of Finance, 2007, 62(3): 1139-1168.

[97] Baker S R, Bloom N, Davis S J. Measuring Economic Policy Uncertainty[J]. Quarterly Journal of Economics, 2016, 131(4): 1593-1636.

[98] Hoberg G, Phillips G. Text-Based Network Industries and Endogenous Product Differentiation[J]. Journal of Political Economy, 2016, 124(5): 1423-1465.

[99] Bharadwaj A, El Sawy O A, Pavlou P A, et al. Digital Business Strategy: Toward a Next Generation of Insights[J]. MIS Quarterly, 2013, 37(2): 471-482.

[100] Vial G. Understanding Digital Transformation: A Review and a Research Agenda[J]. Journal of Strategic Information Systems, 2019, 28(2): 118-144.

[101] Hess T, Matt C, Benlian A, et al. Options for Formulating a Digital Transformation Strategy [J]. MIS Quarterly Executive, 2016, 15(2): 123-139.

[102] Sebastian I M, Ross J W, Beath C, et al. How Big Old Companies Navigate Digital Transformation[J]. MIS Quarterly Executive, 2017, 16(3): 197-213.

[103] Warner K S R, W”ager M. Building Dynamic Capabilities for Digital Transformation: An Ongoing Process of Strategic Renewal[J]. Long Range Planning, 2019, 52(3): 326-349.

[104] Nambisan S, Lyytinen K, Majchrzak A, et al. Digital Innovation Management: Reinventing Innovation Management Research in a Digital World[J]. MIS Quarterly, 2017, 41(1): 223-238.

[105] Teece D J. Business Models and Dynamic Capabilities[J]. Long Range Planning, 2018, 51(1): 40-49.

[106] Goldfarb A, Tucker C. Digital Economics[J]. Journal of Economic Literature, 2019, 57(1): 3-43.

[107] Brynjolfsson E, Hitt L M. Beyond Computation: Information Technology, Organizational Transformation and Business Performance[J]. Journal of Economic Perspectives, 2000, 14(4): 23-48.

[108] Fazzari S M, Hubbard R G, Petersen B C. Financing Constraints and Corporate Investment [J]. Brookings Papers on Economic Activity, 1988, 1988(1): 141-206.

[109] Kaplan S N, Zingales L. Do Investment-Cash Flow Sensitivities Provide Useful Measures of Financing Constraints?[J]. Quarterly Journal of Economics, 1997, 112(1): 169-215.

[110] Almeida H, Campello M. Financial Constraints, Asset Tangibility, and Corporate Investment [J]. Review of Financial Studies, 2007, 20(5): 1429-1460.

[111] Hadlock C J, Pierce J R. New Evidence on Measuring Financial Constraints: Moving Beyond the KZ Index[J]. Review of Financial Studies, 2010, 23(5): 1909-1940.

[112] Whited T M, Wu G. Financial Constraints Risk[J]. Review of Financial Studies, 2006, 19(2): 531-559.

[113] Campello M, Graham J R, Harvey C R. The Real Effects of Financial Constraints: Evidence from a Financial Crisis[J]. Journal of Financial Economics, 2010, 97(3): 470-487.

[114] Ferreira M A, Matos P. The Colors of Investors' Money: The Role of Institutional Investors around the World[J]. Journal of Financial Economics, 2008, 88(3): 499-533.

[115] Aggarwal R, Erel I, Ferreira M, et al. Does Governance Travel around the World? Evidence from Institutional Investors[J]. Journal of Financial Economics, 2011, 100(1): 154-181.

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Patience Fero

2025 年 5 月

Appendix

1 Projects Participated in During the School Period

1. 2023-09-14 (Online, Tencent Meeting): Getting Started as a Research Novice, Speaker: Zhang Jun (Anhui University of Technology).
2. 2023-12-06 (2n311): Golden Opportunity in the Digital Age! Career and Development Prospects in Database Technology, Speaker: Jia Xinhua (Alibaba Cloud).
3. 2023-12-06 (2n311): AI Makes Databases Run "Faster and Further", Speaker: Dai Jian (Alibaba Cloud).
4. 2024-10-23 (Venue not specified): Yongchao Forum: Topic Selection and Writing for Economics and Management Academic Papers, Speaker: Yu Liping (Zhejiang Gongshang University).
5. 2024-11-08 (25-418): Yongchao Forum: Frontier Issues in Science and Technology Development and Reflections on China's Science Evaluation Reform, Speakers: Li Xiaoxuan and Xu Fang (CAS Institutes of Science and Development).
6. 2024-11-14 (2-S343): Yongchao Forum: Frontier Research Perspectives on Digital-Intelligent Transformation in Manufacturing, Speaker: Li Xueling (Jilin University).
7. 2025-10-23 (25-1816): Perspective Taking to Promote Work Design: Research Journey and Inspirations, Speaker: Meng Liang (Shanghai International Studies University).

2 Exploratory Knowledge Spillover Proxy Test

To align the empirical design with recent spillover studies[57-58] while preserving identification discipline, this appendix reports an exploratory proxy test using currently available merged variables. The proxy is leave-one-out peer digital transformation within industry-year cells:

$$\mathrm{PeerDT}_{i,t}^{(\mathrm{ind,loo})} = \frac{\sum_{j\in(\mathrm{ind},t),\, j\neq i} DT_{j,t}}{N_{\mathrm{ind},t} - 1} \tag{0-1}$$

and the exploratory interaction is $\mathrm{MSCI}_{i,t} \times \mathrm{PeerDT}_{i,t}^{(\mathrm{ind,loo})}$.

Table 0-1 Appendix: Exploratory Knowledge Spillover Proxy Regression

Variable	Coeff.	(SE)	*p*-val
MSCI (treatment exposure)	−2.0000	1.8336	0.2754
PeerDT (industry-year leave-one-out mean)	0.7046***	0.0689	0.0000
MSCI × PeerDT	0.2342	0.1481	0.1138
Observations		31,110	
Unique firms (cluster level)		3,921	
R^2		0.3556	

Notes: OLS with firm-clustered standard errors; controls include size, ROA, cashflow/assets, Tobin's Q, independent directors, and ownership HHI, with industry, province, and year fixed effects. The interaction coefficient is positive but statistically insignificant, so this table is interpreted as exploratory context rather than direct mechanism identification.